# 1 About this Open Learning Module

## Target Audience and Prerequisites

This Open Learning module is aimed at new recruits in Great Britain into the role of Signalling Technician on either the installation or maintenance sides. It can also be used to update or refresh the knowledge of experienced technicians who are undergoing re-certification of their competence.

Apart from a basic Secondary education, an ability to understand moderately complex English language and the presentation of information in the form of diagrams, plus a reasonable curiosity and a willingness to learn, the only requirement is that you should have

- successfully completed study of the associated Open Learning Module “Introduction to Railway Signalling”, or have acquired equivalent basic knowledge about the purpose and implementation of railway signalling
- made yourself familiar with the contents of Module 1, Circuit Reading

## Objectives

This module is intended to provide an introduction to the principles of track circuit operation used in railway signalling. It covers

- the basic principles of track circuits
- an introduction to the various types of track circuits
- the DC track circuit
- components of the DC track circuit
- operation of the DC track circuit
- ballast resistance
- DC track configuration
- AC track circuits
- audio frequency track circuits
- high voltage impulse track circuits

## What you will require

Apart from this Open Learning manual the only other material you will require is the Signalling Installation Handbook (GS/IH) and the Track Circuit Handbook (TCH)

## The Results expected from this Module

When you have completed this module, the assessment and been awarded your certificate, you should

- understand the basic principles of track circuits
- have an appreciation of the different types of track circuits
- be able to identify the components of a DC track circuit
- understand how a DC track circuit operates
- understand the effects of ballast resistance
- have an appreciation of the DC track configurations in common use
- have an appreciation of AC, audio frequency and HVI track circuits

## An Introduction to Jargon

You may have wondered about the word **jargon**. When talking to a friend you often talk in a way that you both understand but which a stranger may find very difficult to understand. For example, you might say “the Reds were terrific on Saturday”, meaning the local football team had done well.

Throughout this module you will find yourself introduced to a variety of jargon used on the railways. This will help you to understand not only training modules but also some of the language which signal persons, signalling engineers, signalling technicians and others use when speaking or writing about signalling equipment and systems. You will find the same jargon is used in the many written rules, specifications and drawings you will come across in the course of your work.
The Glossary of Terms (see Section 11) introduces you to some of the variations in the jargon.

## A Safety Warning

This Open Learning module is designed to arouse your curiosity about railway signalling. However, please remember that, until you have undergone proper training and been certified as competent in Personal Track Safety (PTS) and you have been authorised by your supervisor, you are forbidden to go onto any part of the railway that is not open to passengers, unless you are properly supervised and it is an official part of your training.

Study and remember what it says in your Rule Book and the Track Safety Handbook. There is no requirement for you to go onto the railway to successfully complete this course of study.

Of course, your supervisor or appointed Mentor may use their discretion and properly arrange for you to go onto the railway to look at practical examples of the signals, associated equipment and junctions described in this manual and which you have not already had an opportunity to observe in the course of your work and other training.

# 2 Track Circuit Basic Concepts

## 2.1 Introduction

The track circuit is the basis of all modern signalling. It has a number of functions, namely

- to prove that a section of track is clear of a train or vehicles
- to detect the presence of a train or vehicles
- to control signalling equipment

Look at Figure 1, which illustrates the basic principles involved in a track circuit.

*Basic Track Circuit*

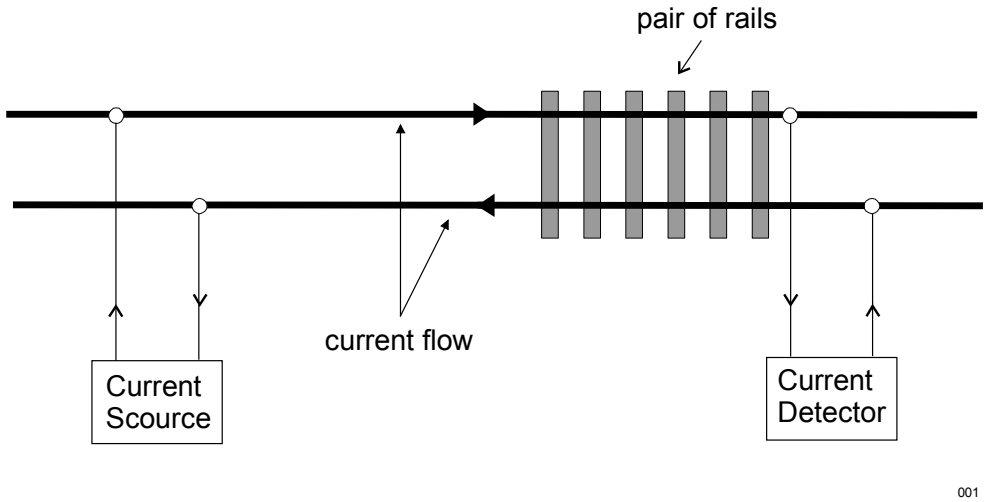



**Figure 1**

The current source causes a current to flow along one of the rails, through the **current detector** and to flow back along the other rail. In the current detector the presence of the current is used to energise a **relay**. When a train or other railway vehicle comes onto the rails its wheels and axles short-circuit the current, which fails to reach the current detector and the relay is de-energised, it can be assumed that a train or other vehicle is **occupying** the rails and the track circuit is said to be **occupied**.

The current used in a track circuit can have many forms. It can be DC, low frequency AC, audio frequency AC or pulsed. A look at the Main Contents list of the Track Circuit Handbook will give you an indication of the variety involved.

The track circuit is essentially a **series circuit**. If any part **breaks** the current path is broken and the track circuit will show **occupied**. The design of the track circuit is such that if a failure occurs such as

- short circuit
- open circuit

the current to the track circuit detector will be cut off. Therefore the unoccupied track circuit has a **fail safe characteristic**, otherwise described as a right side failure. On the other hand, if an occupied track circuit fails to show occupied, that is described as a wrong side failure and could cause an accident to occur.

The rail network consists of numerous track circuits which are **electrically isolated** from each other.

## 2.2 The Train Detection Handbook (TDH)

The Train Detection Handbook provides details of the symbols and abbreviations used directly in connection with a track circuit. If you do not know what a symbol or abbreviation represents, or if you want to check that you have made a correct identification you can refer to the Train Detection Handbook which will be available to you as there is a copy at every depot. Ask your supervisor or mentor where it is located.

The Train Detection Handbook also provides details concerning

- planning and design
- components and installation
- maintenance
- fault finding
- testing and commissioning

Should you require any information concerning track circuits, it is worth referring to the Train Detection Handbook.

## 2.3 Section Summary

This section introduced the basic concept of the track circuit and provided an overview to its function within the railway system.

Before moving on to Section Three make sure that you fully understand all the information which was given in Section Two. If there are any areas which you do not understand, re-read the relevant parts of the text

# 3 The Components of the DC Track Circuit

## 3.1 Introduction

The DC track circuit is the most simple form of track circuit.

The basic components of the DC track circuit are

- rails and bonding
- insulated rail joints
- power supply
- feed resistance or regulating resistance
- track relay

Each of these components will now be examined in greater detail.

*A DC Track Circuit*

Look carefully at figure 2 which gives an example of a DC track circuit layout.

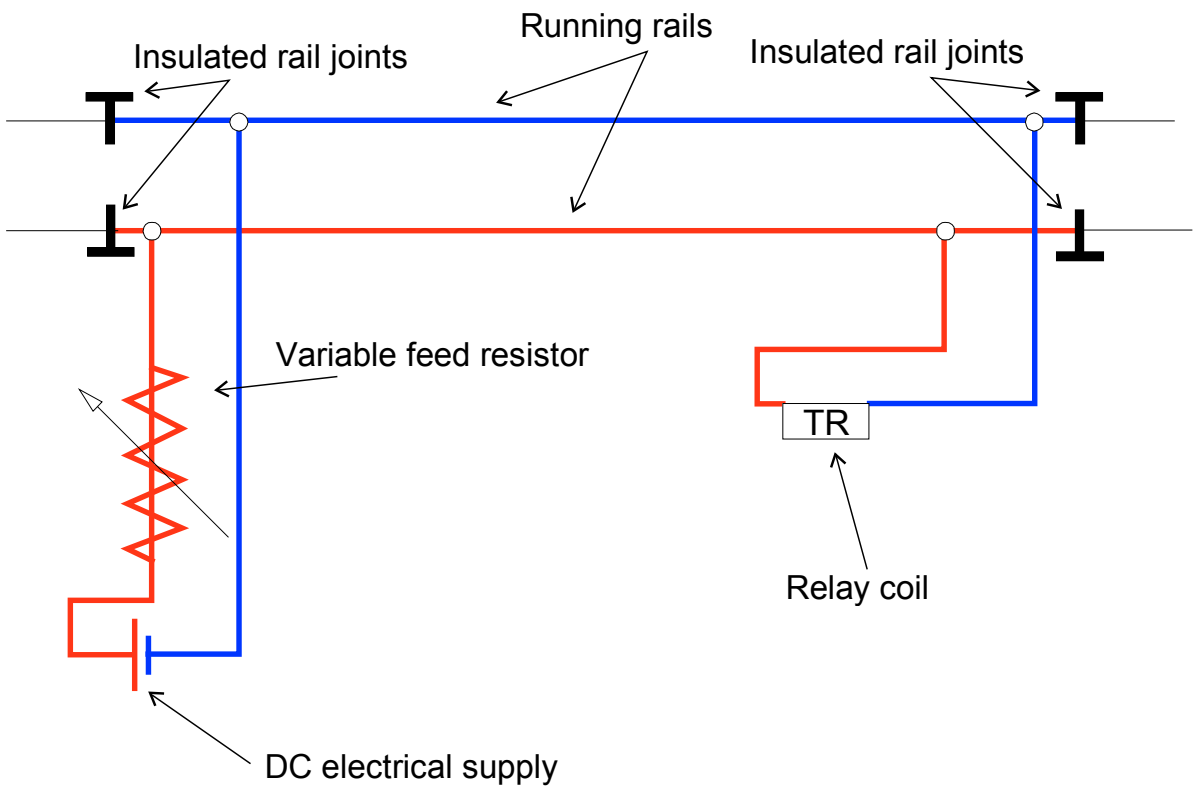



Figure 2

In this case, note that the relay coil is directly energised by the current flowing in the running rails.

## 3.2 Rails and Bonding

The rails provide a **low resistance path** for the current.

### Rails

A power supply is connected to the rails, the

- positive side of the supply is connected to one rail which is called the positive rail
- negative side of the power supply is connected to the other rail and is called the negative rail

Where a positive rail exists a negative rail will be positioned opposite to it.

Figure 3 provides an example of this.

*Example Showing Positive and Negative Rails*

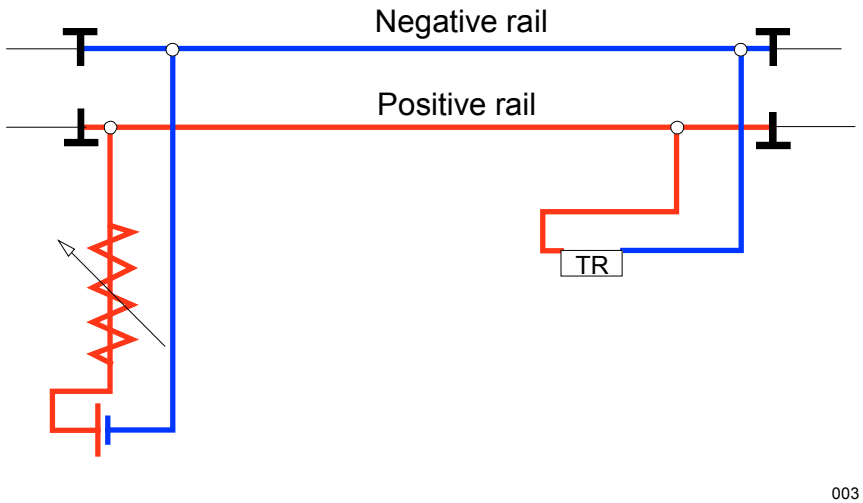



Figure 3

## Bond Wires

Bond wires are provided to ensure a **low resistance path across joints in the rails**. Bond wires are required because fish-plated joints and the connections at points cannot be relied on to conduct electricity.

Bond wires are

- duplicated across each joint or points gap
- inserted into a 9/32" hole on the **four foot side** of the rail
- securely fixed by a tapered channel pin

# 3.3 Insulated Rail Joints

The insulated rail joints (IRJs) isolate one track circuit from the next.

IRJs can be fitted

- the limits of track circuits
- within the track circuit itself

*Typical locations of IRJs*

Figure 4 illustrates the positioning of the IRJs

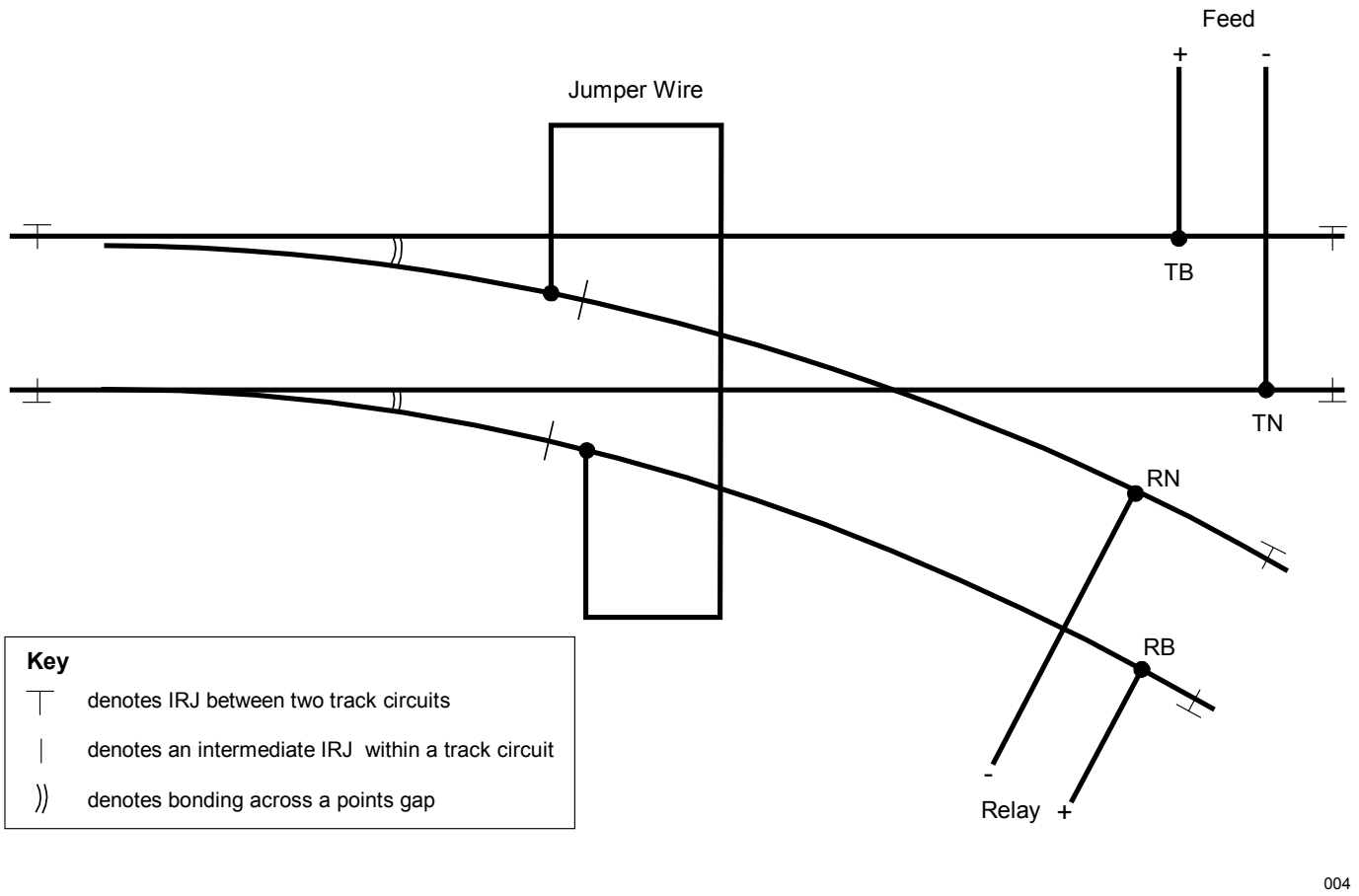



Figure 4

Look carefully at the following list which explains the abbreviations used on figure 4.

TB = Track circuit positive connection at the feed end

TN = Track circuit negative connection at the feed end

RN = Track circuit negative connection at the relay end

RB = Track circuit positive connection at the relay end

Insulated joints may be

- fibre or plastic insulating type with ferrule and insulation pieces
- laminated type
- glued joint type

Figure 5 provides an example of the laminated type of insulated joint.

*Laminated type of Insulated joint*

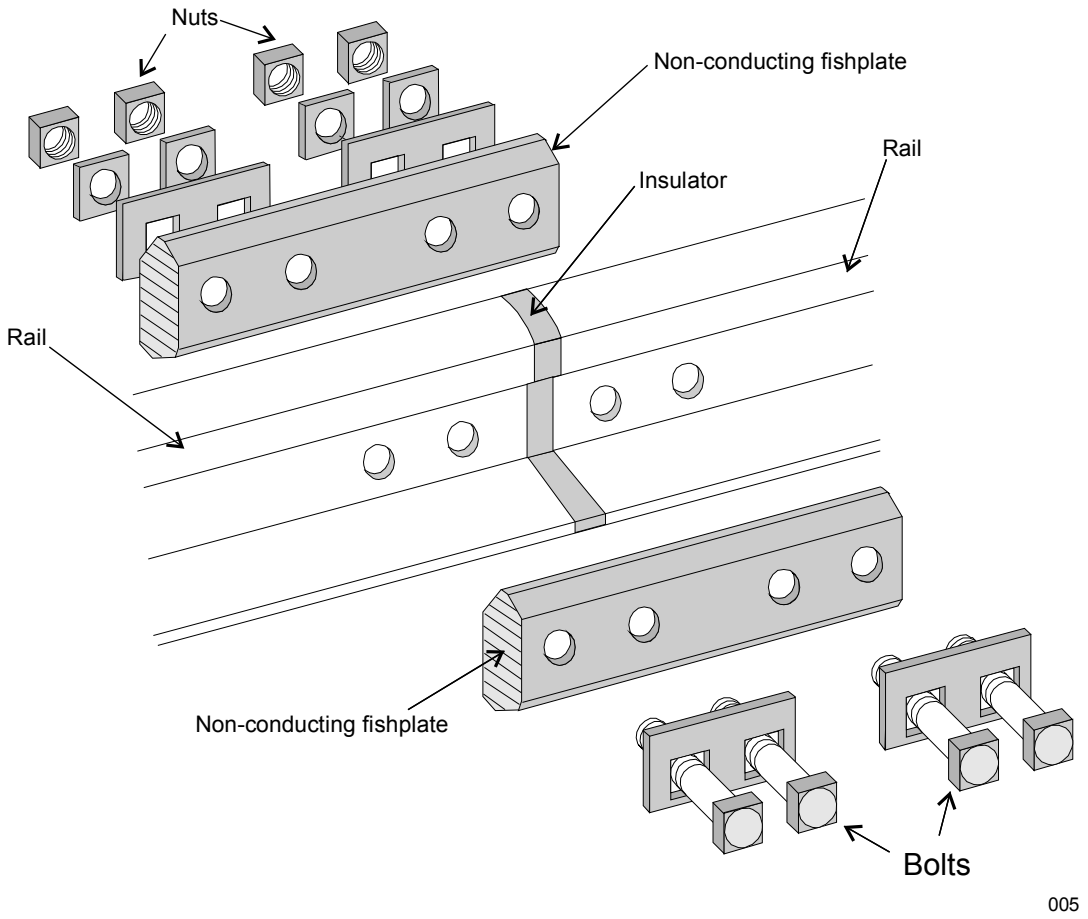



Figure 5

There are several different types of insulated joint but all of them are designed to electrically isolate one rail from the adjacent rail.

## 3.4 Track Circuit Interrupter

Track circuit interrupters are used on track circuited lines which include trap or run-back catch points. These types of points are used to derail any vehicle making an irregular movement over them. It could be a train passing a signal at danger and approaching an unprotected crossing, or wagons running loose from a goods siding, or vehicles running backwards on a hill. In all cases there could be a problem that when they are derailed the vehicles are no longer detected by the track circuit, which then gives an unoccupied indication and a wrong-side failure.

A typical track circuit interrupter is illustrated in figure 6. The two wire connections are arranged to be part of the track circuit and the arrangement is located on the 4 foot side of the stock rail in a position where the flange of a train wheel making an irregular movement will break off the head portion at its narrow neck. The track circuit is broken and indicates occupied until the interrupter is repaired.

*Track Circuit Interrupter*

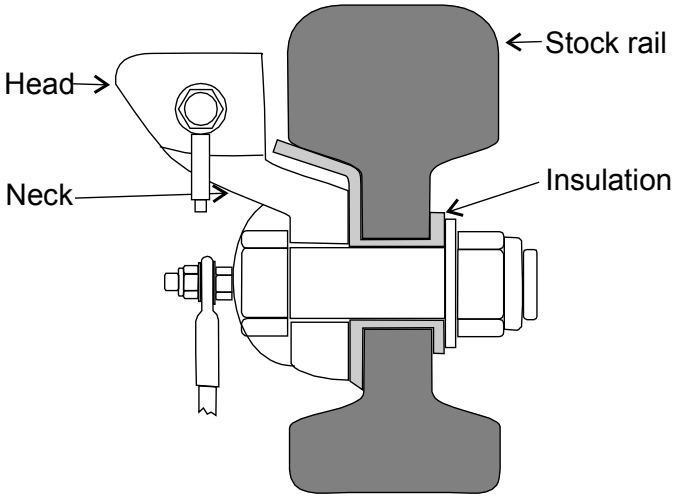



Figure 6

The positive feed to the track circuit arrives at the head of the interrupter and leaves at the bottom to go to the TB connection on the track circuit.

The interrupter is bolted to the negative rail. This is in case the head of the interrupter is knocked off and makes contact with the rail.

If this happens the resulting short circuit does not prevent the track circuit showing occupied.

## 3.5 Power Supply

The power supply provides the **voltage** between the rails. In certain areas track circuits are **battery fed.**
Track circuits in power signalling areas obtain their supply from a specially designed transformer/rectifier set which converts AC to DC.

## 3.6 Feed Resistance

The feed resistance

- regulates the flow of current to the track circuit
- prevents a short circuit on the battery or feed set when the track circuit is occupied

This will be examined in greater detail in Section Four.

The regulating resistor may be either an

- adjustable unit separate from the power supply
- integral part of the power supply unit

## 3.7 Track Relay

The types of relay used in DC track circuits are designed to meet specific requirements.

In addition to the common details which are found on a relay label there may be information concerning two values, namely

- pick up
- drop away

These two terms may be referred to as

- pick up current or voltage
- drop away current or voltage

They determine the effectiveness of the relay in distinguishing between the track not occupied and the track occupied.

These values and other requirements will be examined in greater detail in the next section.

## 3.8 Section Summary

This section has looked at the component parts which make up the DC track circuit.

Now complete Self-Assessment Question 1 before proceeding to Section Four.

## Self-Assessment Question 1

Correctly identify the following component parts of a track circuit shown in figure 7.

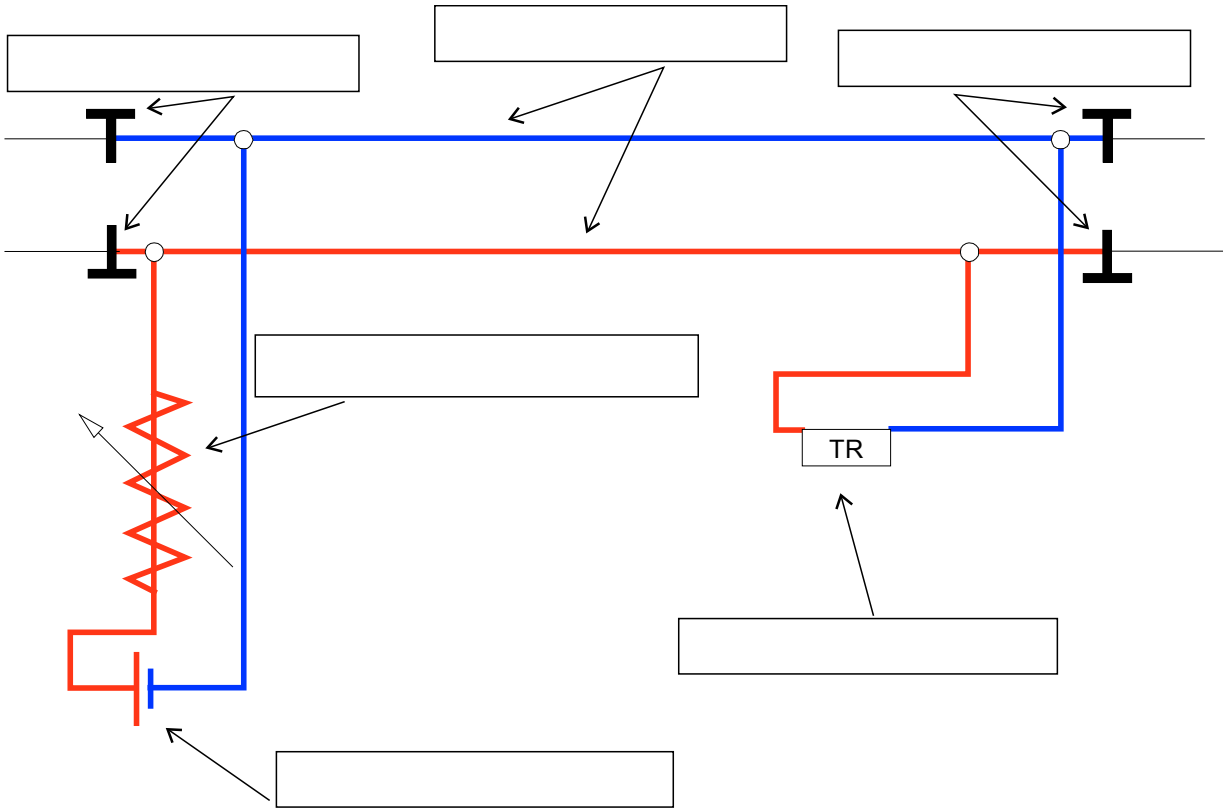



Figure 7

Now check your answer with the one given in the Answers Section. Before moving onto Section Four make sure that you fully understand all the information given in Section Three. If there are any areas which you do not understand re-read the relevant parts of the text.

# 4 Operation of The DC Track Circuit

## 4.1 Introduction

In Section Three you examined the component parts of the DC track circuit. At this stage you will now consider the **operation** of the DC track circuit.

There are two basic states for the track circuit

- track circuit unoccupied (clear)
- track circuit occupied

Each of these basic states will now be examined in greater detail.

## 4.2 Track Circuit Unoccupied

Look carefully at figure 8 which shows the **track circuit** unoccupied.

*Track Circuit Unoccupied*

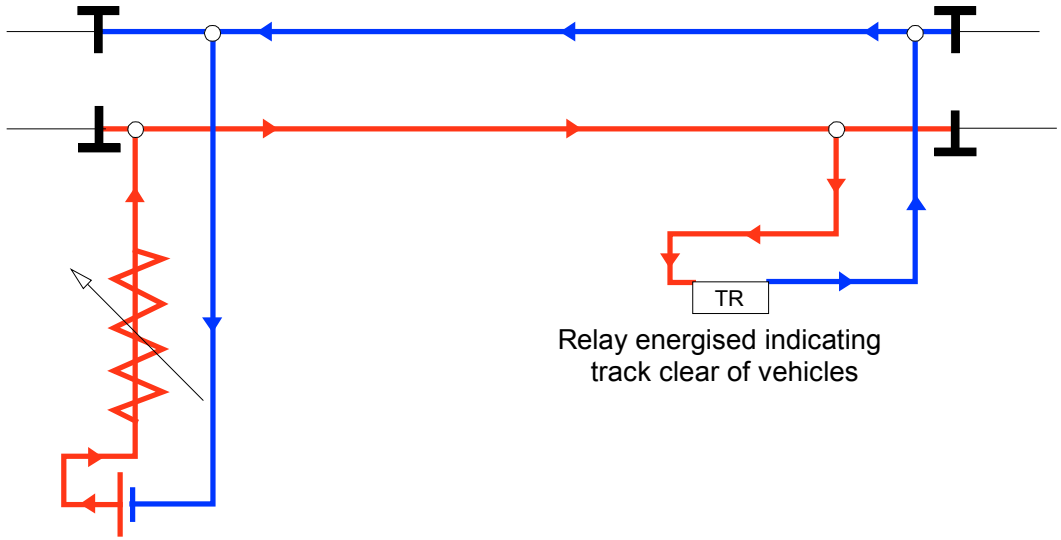



Figure 8

In figure 8 it can be seen that the running rails of a section of a line are electrically isolated by the provision of four insulated rail joints

- one end of the section is connected through a variable resistance to a battery
- the other end of the section is connected to a track relay

Current flows from the battery through the variable feed resistance which regulates the flow of current via one rail to the track relay. The current continues to flow through the relay winding from where it returns through the other rail to the negative of the battery.

This current flow energises the track relay which makes its front contacts.

This information can be simplified as follows, from figure 8 it can be seen that when the track circuit is **clear** the flow of current is

- from the positive battery or feed supply through the feed resistor to the positive rail
- from the positive rail through the track relay coil and back to the negative rail
- from the negative rail to the battery negative terminal

## Self-Assessment Question 2

State the two basic conditions of the track circuit.

Now check your answer with the one given in the Answers Section.

## 4.3

# Track Circuit Occupied

Look carefully at figure 9 which shows the track circuit occupied.

*Track Circuit Occupied*

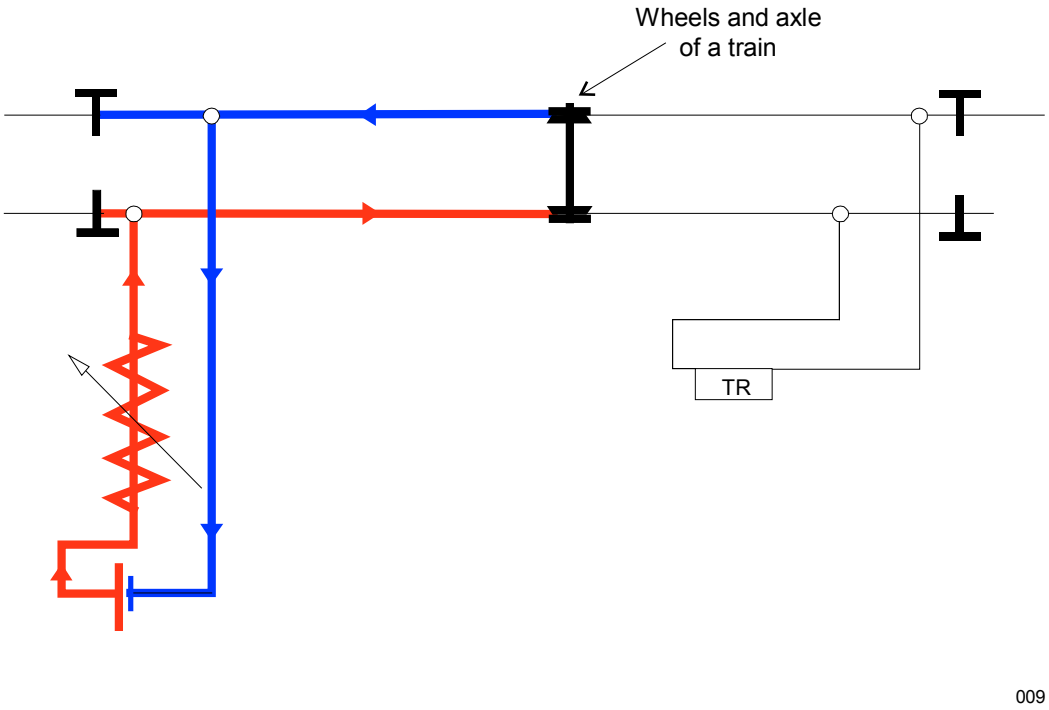



Figure 9

Figure 9 shows the same track circuit illustrated in figure 8 but in this case the track circuit is occupied by a train.

Current flows from the battery through the variable resistance to the **rail**. The two rails are now bridged by the **wheels** and the **axles** of the train.

The electrical resistance of the train is **negligible** compared to that of the relay coil of the track relay.
The current chooses the line of **least resistance** and flows through the wheels and axles rather than through the relay coil.

The relay is effectively **by-passed**. The proper term for this is **shunted**.
Practically **no current** passes through the relay coil. This means that when the track is occupied

- the relay becomes electrically de-energised
- the front contacts break

This information can be simplified as follows, when the track circuit is occupied

- the current flow is from the positive battery or power supply through the feed resistor to the positive rail
- the train wheels and axles make a short circuit between the positive and negative rails. As this offers the least resistance path to the current, the majority of the current flows through the wheels and axles to the negative rail and back to the battery or power supply
- some current still flows through the rails to the track relay but it will be negligible
- the relay is therefore de-energised because the current has been shunted away

The track circuit is limited in its **workable length** by two factors, namely

- the resistance of the rails themselves and the jumper bonds
- leakage of current from one rail to the other through the
  - rail fixings
  - sleepers
  - ballast

You will now look at the first of these factors more closely.

## 4.4 Bonding

The Track Circuit Handbook provides detailed information concerning bonding. It may be useful to locate the relevant information and check the conditions which are laid down concerning bonding. If you are unsure of any details concerning the use of bonding, always check the information provided in the Track Circuit Handbook.

It has already been mentioned earlier in the text that there are **two** basic types of bonding, namely

- series
- parallel

Each of these types of bonding will now be examined in greater detail.

## Series Bonding

With series bonding each rail and bond are connected in series.

Look carefully at figure 10 which provides an example of series bonding.

*Typical Example of Series Bonding*

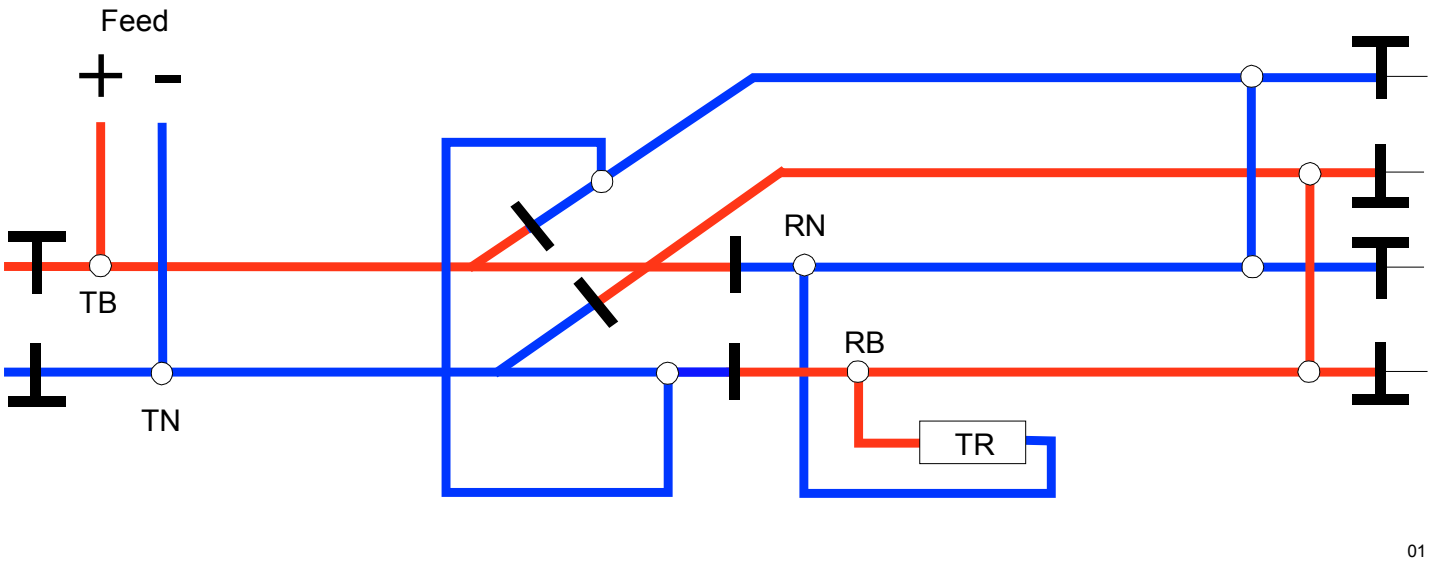



Figure 10

If a break occurs anywhere in the circuit, the track circuit will de-energise.

# Parallel Bonding

It is normally necessary to bond all elements of the track circuit in series. This will ensure that the track circuit will fail safe should a bonding disconnection or a broken rail occur.

In certain circumstances it is physically impossible to achieve total series bonding to both rails. If parallel bonding is used, steps must be taken to ensure that a disconnection will not result in a train or vehicle going undetected. For this reason the Track Circuit Handbook refers to parallel bonding as a non-preferred method.

Figure 11 shows the track circuit of figure 10 converted to show parallel bonding.

*Example of Non-preferred Parallel Bonding*

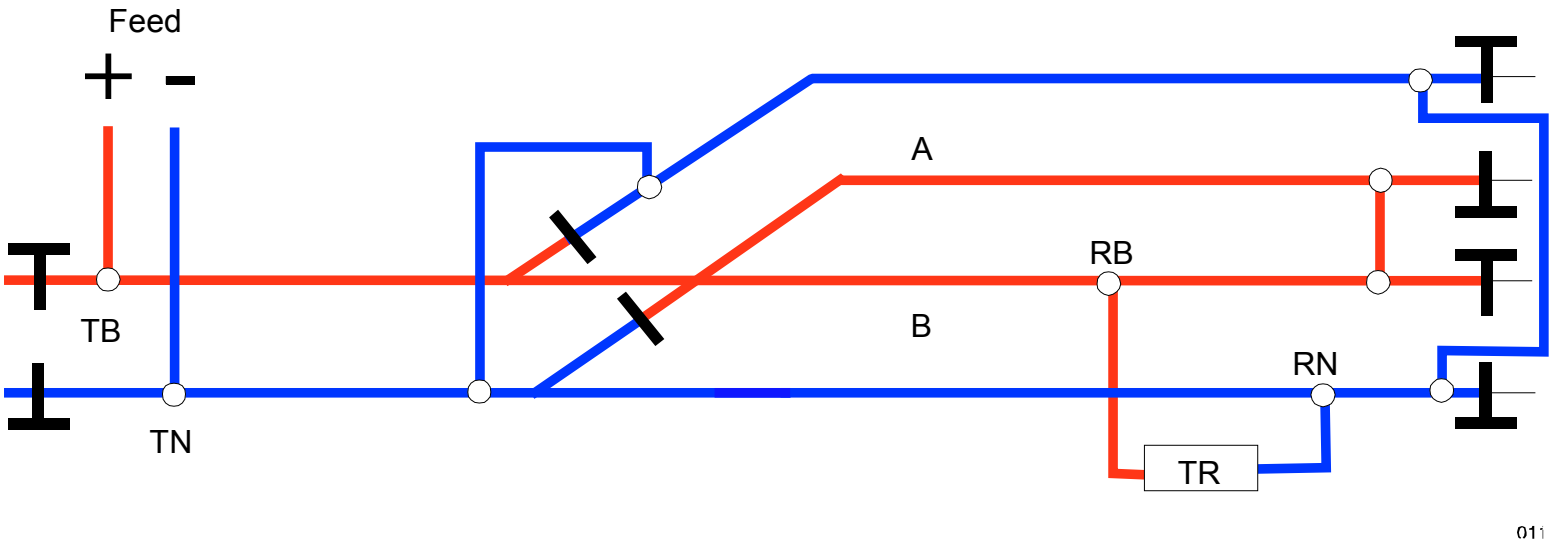



Figure 11

Normally the track relay will show occupied if a train is in section A, but if the jumper wire or a rail in section A is broken the presence if that train will not be detected and the track circuit will not fail safe (a wrong side failure).

*Yellow Jumper Wires*

To guard against that possibility, it is normal practice to provide multiple jumper wires connected to a variety of points on the rails forming the track circuit (as shown at the right hand side of figure 11). Where multiple jumper wires are used they are identified by having yellow sheaths or yellow sleeves at their terminations. This is done to mark them out for special maintenance attention.

Parallel bonded track circuits will operate satisfactorily providing that all the component parts are intact. However, should a component failure occur a wrong side failure may be created. The Track Circuit Handbook section B20 details the necessary safeguards required if parallel bonding is used.

## Ballast Resistance

Ballast resistance is the resistance measured between the two rails with the supply and relay disconnected. The effect of ballast resistance will be examined in greater detail in Section Five.

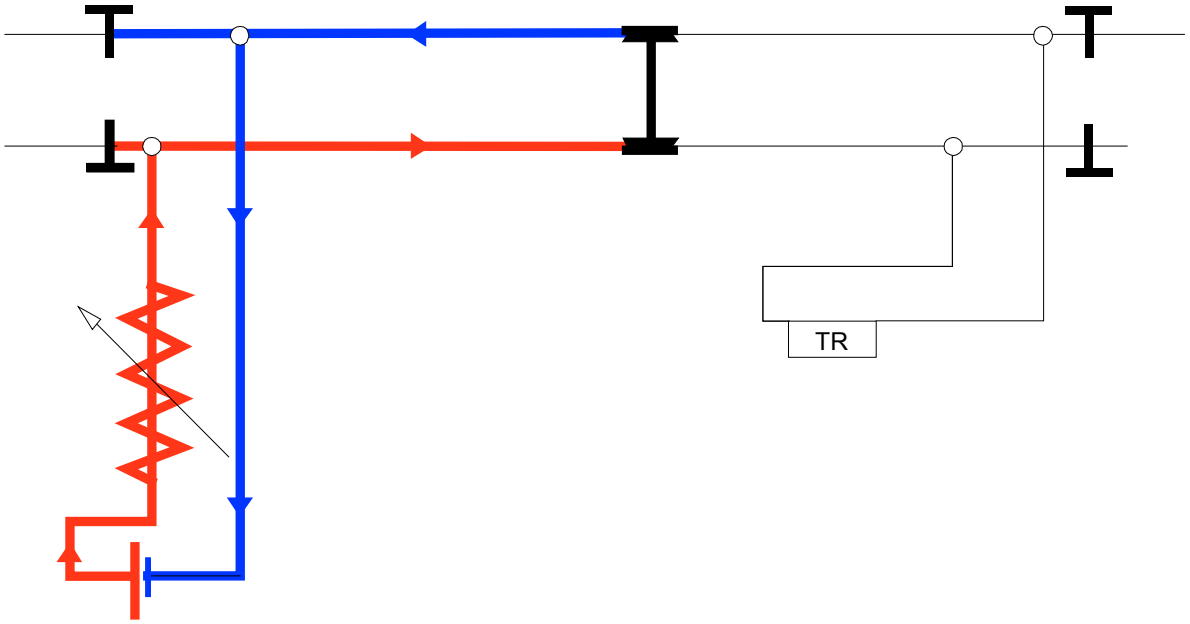
## 4.5 Section Summary

In this Section you have looked at the operation of the DC track circuit.

Now complete Self-Assessment Question 3 before proceeding to Section Four.

## Self-Assessment Question 3

Look carefully at figure 12 and identify the state indicated by the track circuit.




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Figure 12

Now check your answer with the one given in the Answers Section. Before moving on to Section Five make sure that you fully understand all the information which was given in Section Four. If there are any areas which you do not understand, re-read the relevant parts of the text.

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# 5 Ballast Resistance

## 5.1 Introduction

In Section Four it was stated that ballast resistance is the resistance measured between two rails with the supply and the relay disconnected.

Ballast resistance may vary from

- 50 ohms per 305 metres (1000 feet) for well ballasted dry track with insulated rail fixings
- 0.5 ohms per 305 metres in wet conditions, or when the ballast is dirty

Ballast resistance is the **electrical resistance** of all the material between the two rails including

- rail fixings
- sleepers
- the ballast

An example of ballast resistance is shown in figure 13.

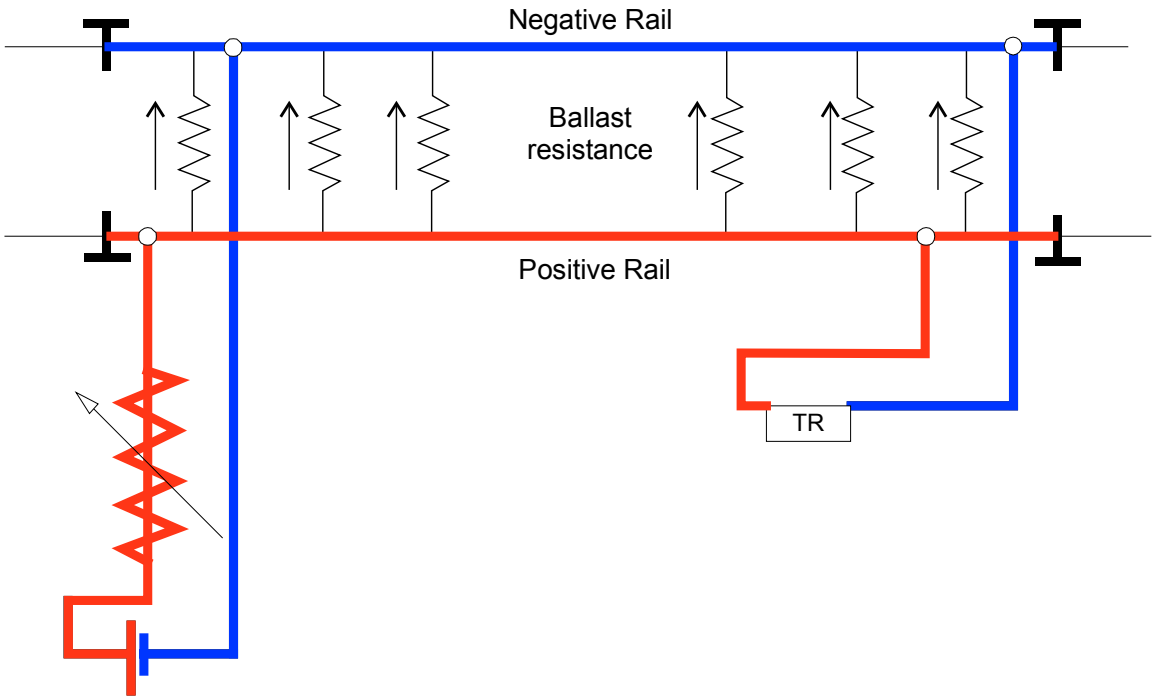



Figure 13

Figure 13 illustrates the leakage of current between the

- positive rail
- negative rail

The electrical equivalent of figure 13 is shown in figure 14.

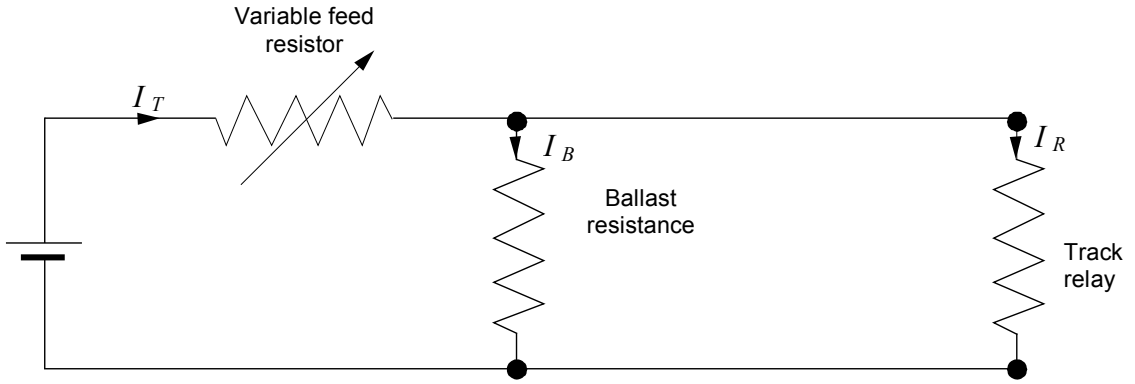



Figure 14

It can be seen in figure 14 that some of the current from the supply will flow through the ballast resistance

- the current flowing through the ballast resistance is represented by $I_B$
- the current flowing through the relay is represented by $I_R$
- the total current being supplied by the battery ( $I_B$ and $I_R$ ) is represented by $I_T$

The ballast resistance is distributed over the full length of the track.

In normal operation the circuit should energise the relay when there is **no train** on the track circuit. If the ballast resistance in figure 14 is low enough, the current energising the coil can be reduced to a level where the relay becomes **de-energised**. This would indicate that there is a train on the track when in fact the track circuit is clear.

This is one of the reasons why the **variable feed resistor** is connected to the supply. It can be adjusted to take account of the ballast resistance and ensure correct operation of the track circuit.

It is important to recognise that ballast resistance constantly varies with **weather conditions**.

The basic conditions are

- wet
- dry
- frosty

The effects of each of these weather conditions will now be examined in greater detail.

## Wet

In wet weather the ballast resistance is **low**. This means that **more** current is shunted away from the relay.

## Dry

In dry weather conditions ballast resistance is **high**. This means that **less** current is shunted away from the relay.

## Frosty

When the ballast is frozen the moisture present is turned to ice which has a much higher resistance. As a consequence, frosty conditions can give a high ballast resistance similar to dry conditions.

## 5.2 Drop Shunt Value

The **drop shunt value** is that value of resistance which must be connected across the track (at the track relay end) to just cause the track circuit to indicate occupied.

In order for the track circuit to operate correctly, the optimum drop shunt value must be obtained. To achieve this the value of the rail voltage must be carefully adjusted.

The drop shunt value can be established by the **drop shunt test**.

The drop shunt value is also known by the terms

- train shunt value
- drop away shunt value

## 5.3 Drop Shunt Test

Under average weather conditions an optimum drop shunt value of 1 ohm must be achieved.

The minimum permissible drop shunt value is 0.5 ohm. Any less than this on a DC track circuit must be investigated immediately as a **wrong side** failure could occur.

The shunt box should be set at **maximum resistance** and then connected across the rails. The resistance should then be **reduced gradually** until the **front** contacts of the **track relay** are seen to break. The detailed procedure to establish this value is provided in the Track Circuit Handbook.

This test should be repeated three times so that an average value can be obtained, unless the first two tests provide the same result.

To alter the value of the drop shunt, the **variable feed resistor** must be adjusted.

**Increasing** the value of the variable feed resistor will

- **reduce** the rail voltage
- **increase** the drop shunt value

**Reducing** the variable feed resistor will

- **increase** the rail voltage
- **reduce** the drop shunt value

The drop shunt test is carried out to ensure that the track circuit will

- always de-energise when a train occupies the track circuit
- that the track circuit is reliable in all weather conditions.

## 5.4 Section Summary

This section has looked at

- ballast resistance
- drop shunt value
- the drop shunt test

Now complete Self-Assessment Question 4 before proceeding to Section Six.

## Self-Assessment Question 4

Table 1 shows a set of figures produced on an average day. Complete the table by stating how the values will be affected by weather conditions. You should place the terms

- more
- less
- same

within the empty boxes provided in Table 1.

| Ballast condition | Ballast resistance | Drop Shunt Value | Voltage at relay |
|---|---|---|---|
| Average | 10 ohms | 0.7 ohms | 0.6 ohms |
| Wet | | | |
| Dry | | | |
| Frosty | | | |

Table 1

Now check your answer with the one given in the Answers Section. Before moving on to Section Six make sure that you fully understand all the information which was given in Section Five. If there are any areas which you do not understand, re-read the relevant parts of the text.

# 6 DC Track Configuration

## 6.1 Introduction

There are **three** configurations of track circuits which are in common use, these are

- double rail operation
- single rail operation
- common rail

Each of these configurations will now be examined in greater detail.

## 6.2 Double Rail Operation

The tracks which have been examined up to this stage have all used **double rail operation**. In this mode of operation **both rails** belong to the track circuit **alone**. This means that both are **insulated** from any adjoining rails.

Look carefully at figure 15 which shows an example of double rail operation.

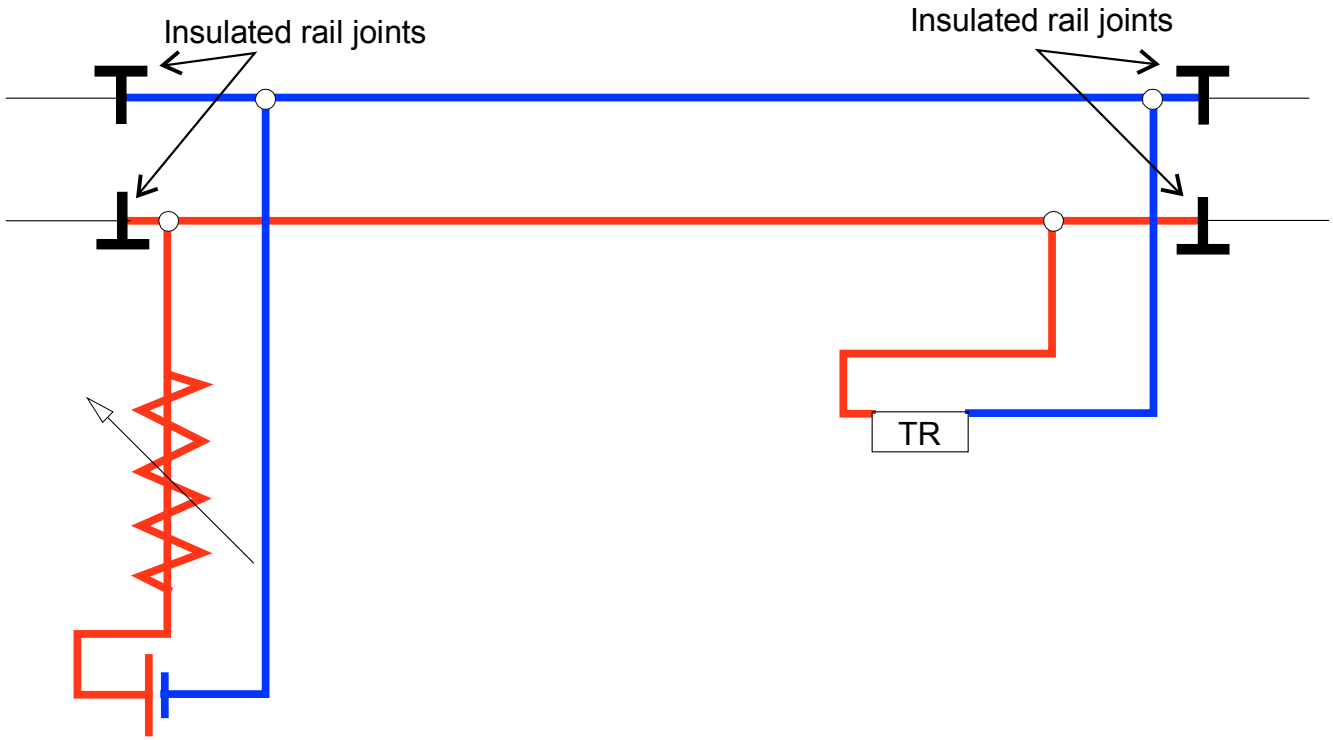



Figure 15

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From figure 12 you can see that insulated rail joints are used on **both rails**, this means that the

- positive rails are insulated
- negative rails are insulated

## 6.3 Single Rail Operation

Single rail operation is employed where **overhead electrical traction** is used. This means that one of the rails is **shared** as a **return path** for the electric traction current.

No insulated rail joints are used on the shared rail. Extra bonding is used on this rail for the electric traction return. The rail which is shared for the electric traction return is normally termed the **continuous traction return rail.** The other rail is generally referred to as the **signalling rail**.

*Parallel bonded return paths*

Where traction return rails are used, it is standard practice to use **parallel bonded return paths** for the traction return. In the event of a break in the continuity of this bonding there will be an alternative path. This means that there is less chance of a dead section in the traction return occurring, which could lead to a wrong side failure.

*Yellow Bonding Cables*

When parallel bonded return paths are used for the traction return a thick cable is used to withstand the higher currents involved and the cables are clearly identified for special maintenance attention by having either a yellow sheath or yellow sleeves at their ends.

*Single Rail Operation*

Look carefully at figure 16 which shows an example of single rail operation.

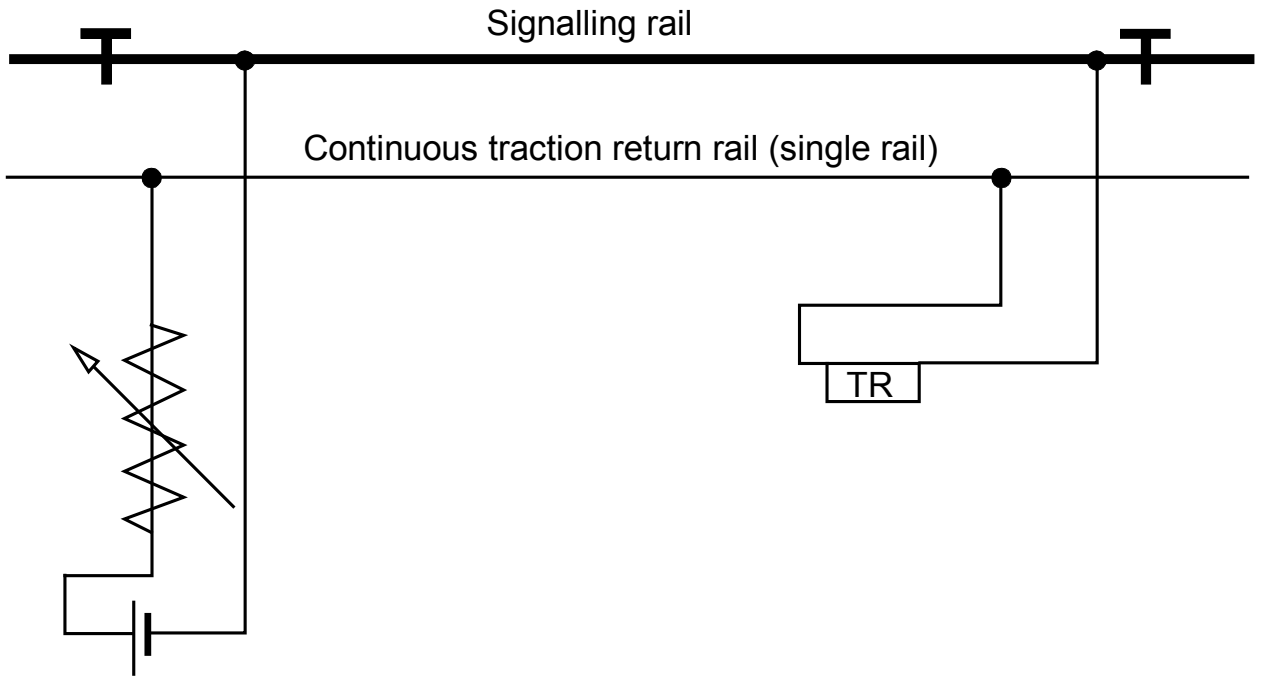



Figure 16 017

## 6.4 Common Rail Operation

Common rail operation is similar to single rail operation where only one rail (the signalling rail) has insulated rail joints to separate the track circuits. The other rail (the common rail) is electrically continuous but not used for electric traction return. This is illustrated in figure 17.

*Common Rail Operation*

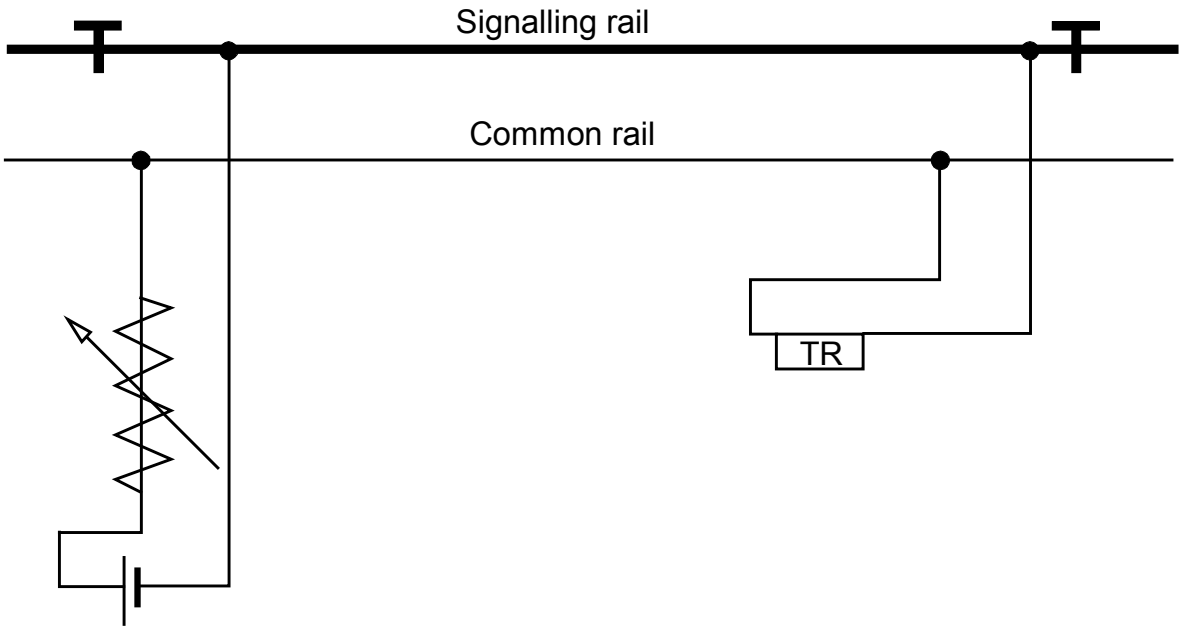


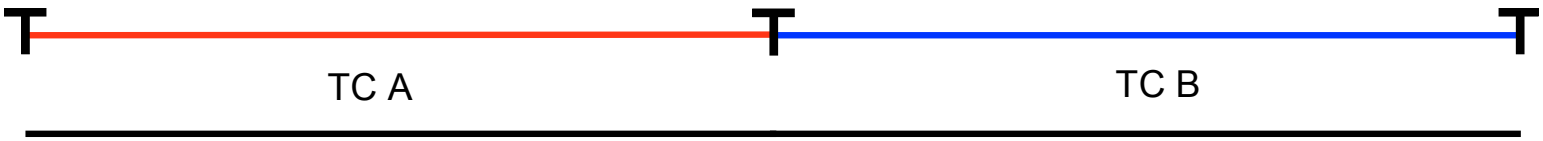

Figure 17 018

As with single rail operation, any bond wires required to maintain the continuity of the common rail are paralleled to reduce the possibility of a bonding break causing a wrong side failure. Yellow bonding wires are used where a parallel bonding exists.

## 6.5 Staggering

Look carefully at the series of drawings shown in figures 18 to 20, these will be used to explain the process of staggering.

*Staggering*




01

Figure 18

The basis of the configuration shown in figure 18 is that the polarity on the rails of the adjacent track circuits is alternated, as shown by TC A and TC B.

Look carefully at figures 19 and 20.

*Illustration of Track Circuits with no Stagger*

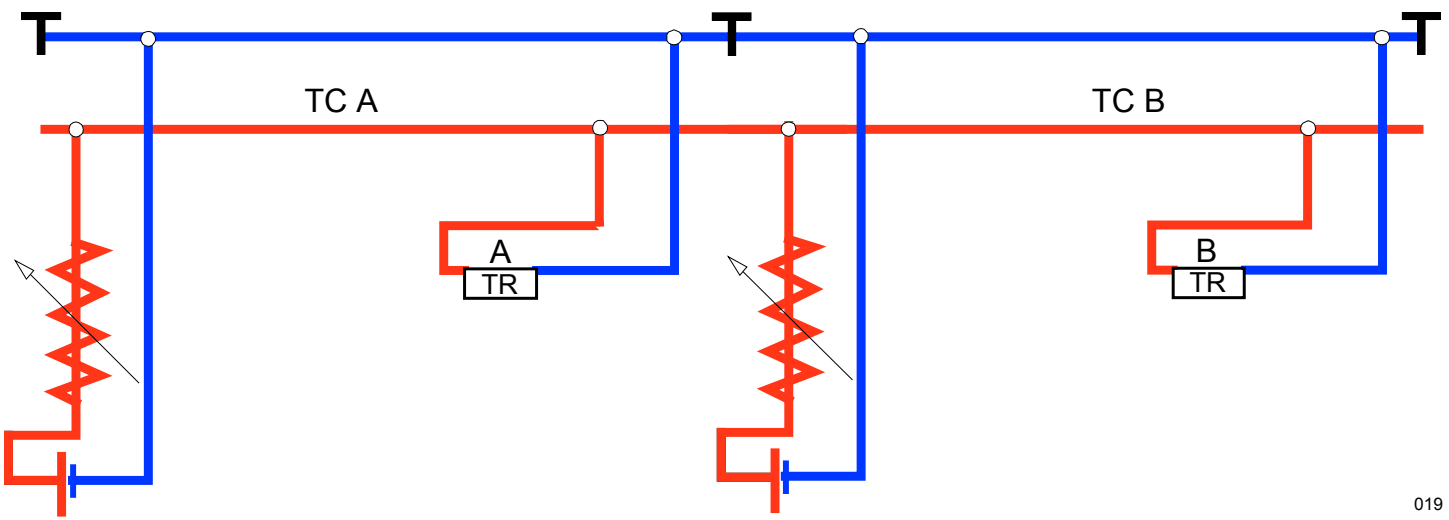
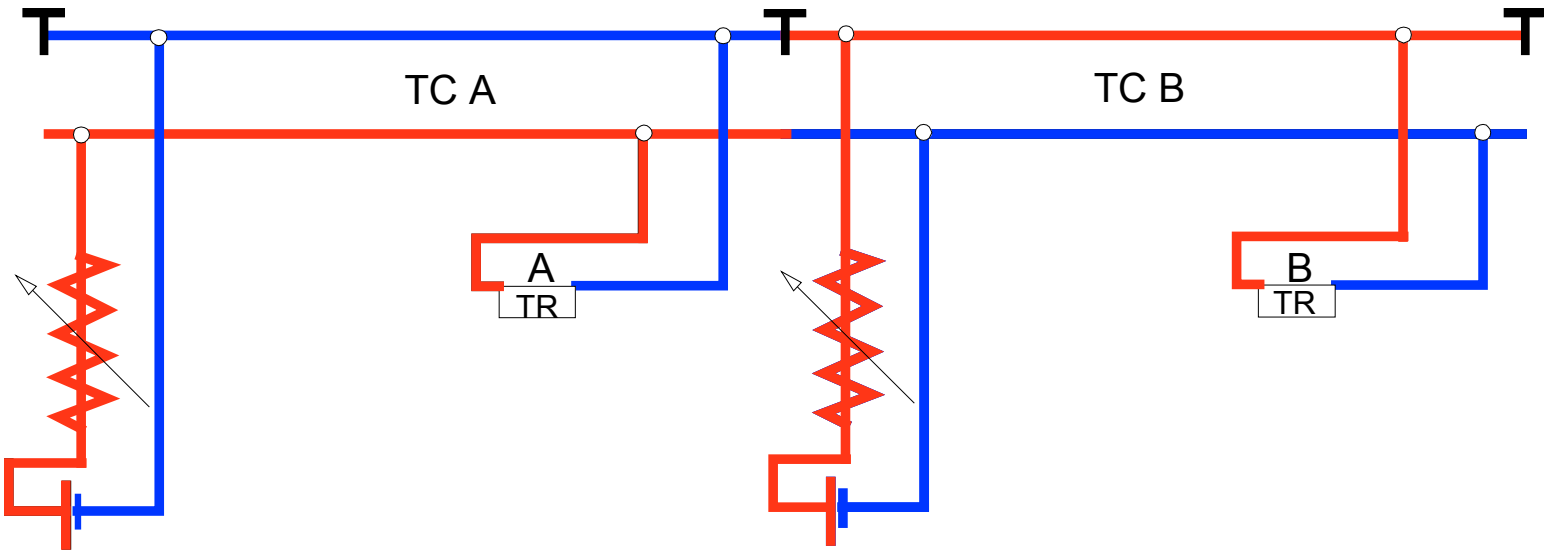



Figure 19

In figure 19 the track circuits TC A and TC B are **not staggered**. If the insulated rail joint between the two track circuits fails the two track feeds **assist** each other. This can result in a **wrong side failure**.

*Illustration of Track Circuits with Stagger*




020

Figure 20

In this case, in the event that the insulated rail joint between the track circuits fails, the effect is

- the supply currents oppose each other
- both relays release
- the track circuits fail right side

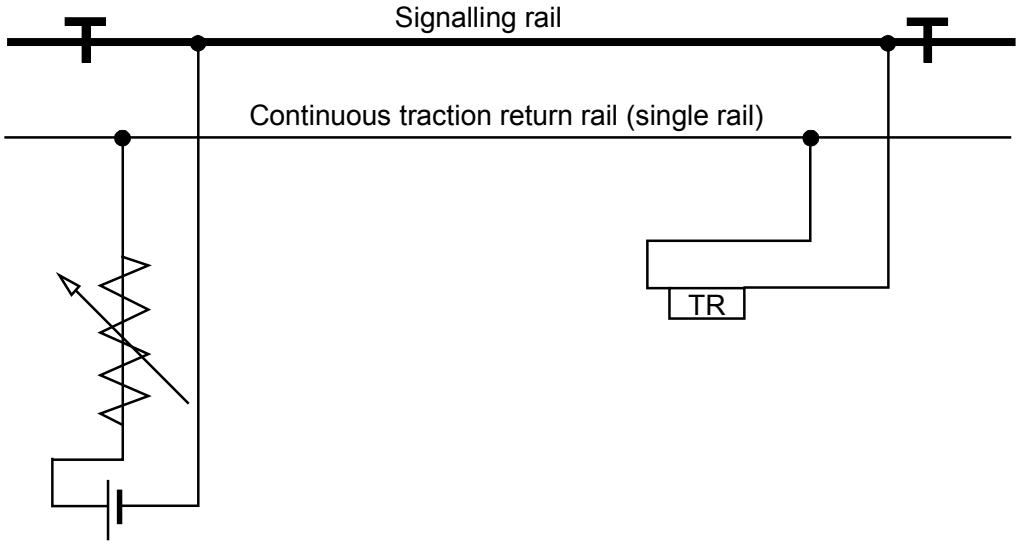
## 6.6 Section Summary

This section has looked at the different types of DC track circuit configurations.

Now complete Self-Assessment Question 5 before proceeding to Section Seven.

## Self-Assessment Question 5

State the type of DC track circuit configuration shown in figure 21.




017

Figure 21

Now check your answer with the one given in the Answers Section. Before moving on to Section Seven make sure that you fully understand all the information which was given in Section Six. If there are any areas which you do not understand, re-read the relevant parts of the text.

# 7 Other Types of Track Circuit

## 7.1 Introduction

So far this module has introduced you to DC track circuits. In this section you will look at some of the other basic types of track circuits.

*Types of Track Circuit*

- AC track circuits
- audio frequency track circuits
- high voltage impulse track circuits

## 7.2 AC Track Circuits

These were first brought into use with the introduction of DC traction where the traction return currents in the rails prevented DC track circuit relays from dropping and caused wrong side failures. The design of AC track circuits is not vastly different from the DC track circuits. The only fundamental differences are:

- the low DC voltage source is replaced by a low AC voltage source
- the DC track relay is replaced by an AC vane relay.

*50 Hz/75 Hz/83.3 Hz supplies*

Initially the AC supply was taken from the standard 50 Hz mains, but with the introduction of AC traction the use of 75 Hz and 83.3 Hz supplies has become more common. These frequencies are not produced by the harmonics of 50 Hz.

Figure 22 shows the basic arrangement of an AC track circuit when single rail operation is employed.

*AC Track Circuit with Single Traction Return Rail.*

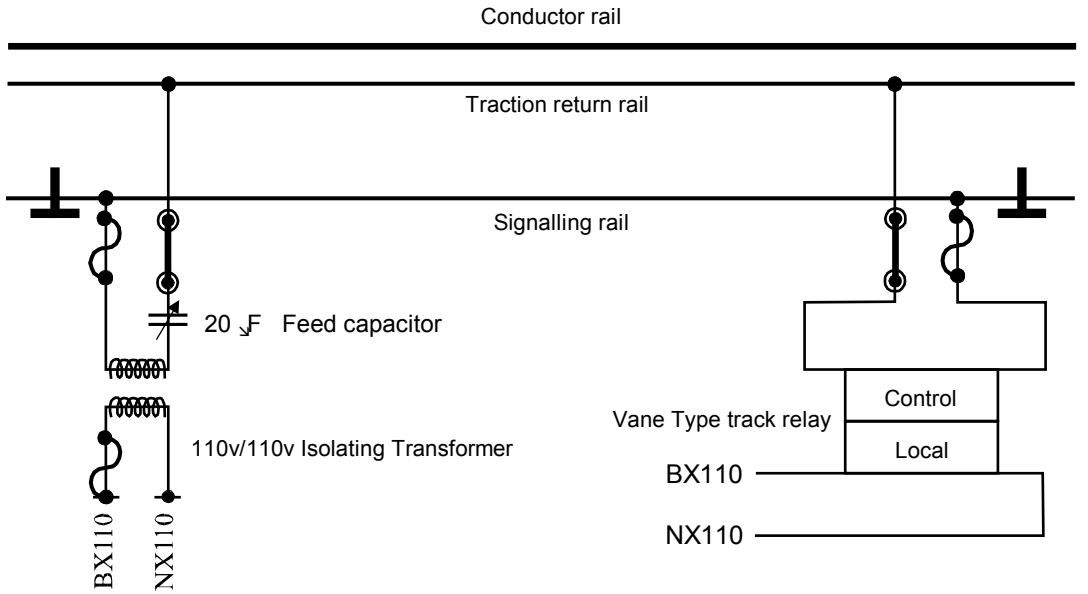



**Figure 22**

027

Note that a variable feed capacitor is used in lieu of a feed resistor to provide a DC block and the AC phase shift required to operate the vane relay. The supply to the local coil on the vane relay has to come from the same source as the energisation to the control coil.

## Double Rail AC Track Circuit in DC Traction Areas

Single traction return rails are practical when the traffic is light. However, when the traction currents are heavy or the distances between the substations are long, the voltage drop in a single return rail becomes excessive and both running rails have to be used for the traction return current. In this case, impedance bonds are used at each end of the track circuit to separate the AC signalling current from the DC traction current. Figure 23 shows the basic arrangement.

*AC Track Circuit with Double Traction Return Rail*

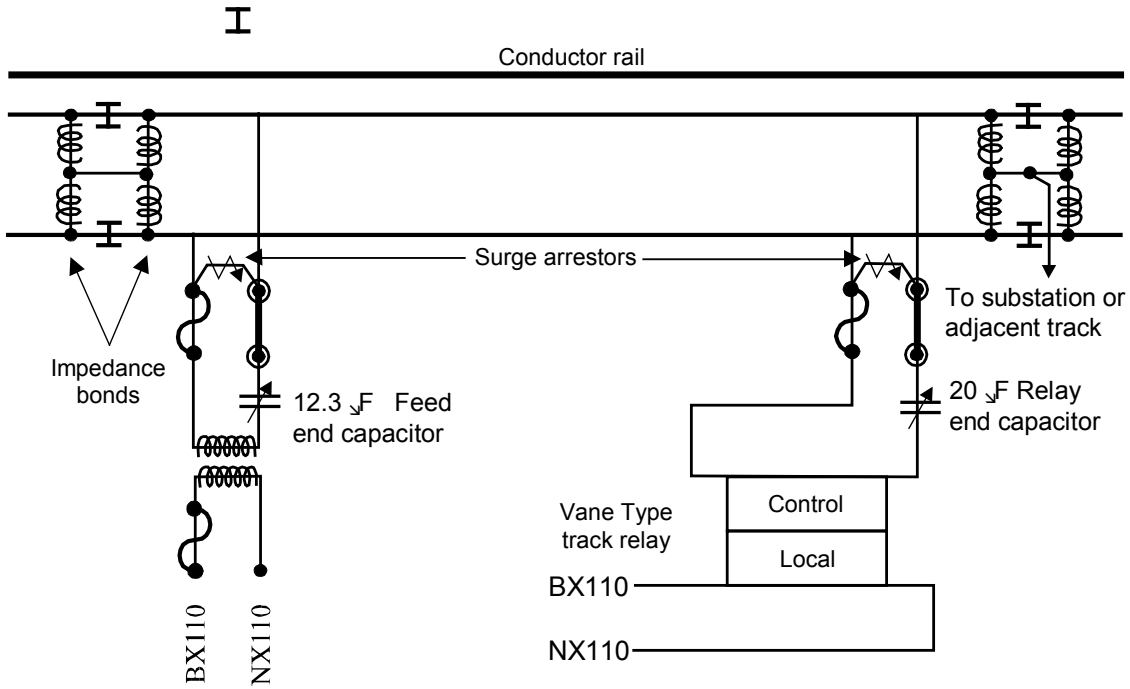



**Figure 23**

028

*Arrangement of Simple Impedance Bond*

The operation of the impedance bond is illustrated in Figure 24

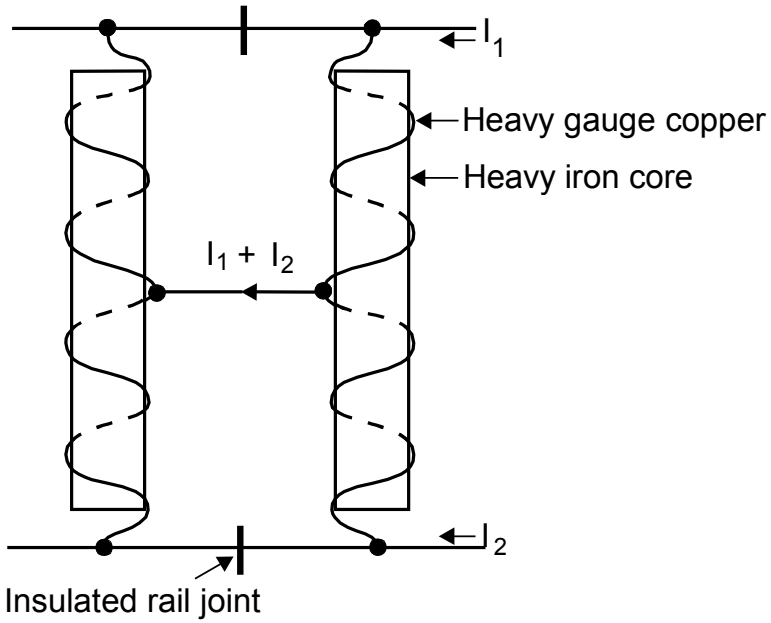



**Figure 24** 029

Each impedance bond consists of a heavy iron core wound with two coils of heavy gauge copper which are connected in series across the rails. The centre point is connected to the substation busbar or to the centre point of the next impedance bond. The DC traction return currents in each rail ($I_1$ and $I_2$) flow through the centre point connection and magnetise the iron core in opposite directions. If the coils are perfectly symmetrical and $1_1$ and $I_2$, the net DC induced flux is zero. In practice that perfect symmetry is never achieved and a large air gap is provided in the iron core to ensure that typically it does not saturate until the current imbalance exceeds 20%. The AC track circuit current flows in series through the two coils. Provided the core is not saturated a significant impedance is offered for the AC.

## Resonated Impedance Bond

The need to maintain a large air gap in the iron core of the impedance bond in order to tolerate a large DC traction current imbalance means that the impedance offered to the track circuit current is quite low, typically 0.4 ohm at 50 Hz. This in turn means that the drop shunt resistance is also too low to be practical, particularly with light trains or infrequently used lines where rust and other line surface contamination is a problem.

The performance of the impedance bond from a signalling point of view can be substantially improved by resonating the traction coil. This is done by connection of a variable capacitor in parallel and adjusting it to obtain resonance at the signalling frequency. In practice the capacitor is connected across a step-up secondary winding. Typical turns ratios of 50:1 are used. The main advantage is this reduces the capacitance required to a practical value. A typical arrangement of the resonated impedance bond is shown is figure 25. Note that the surge arrestor is now transferred to the secondary winding, where its main purpose is to ensure that the maximum working voltage of the capacitor is not exceeded.

*Resonated Impedance Bond*

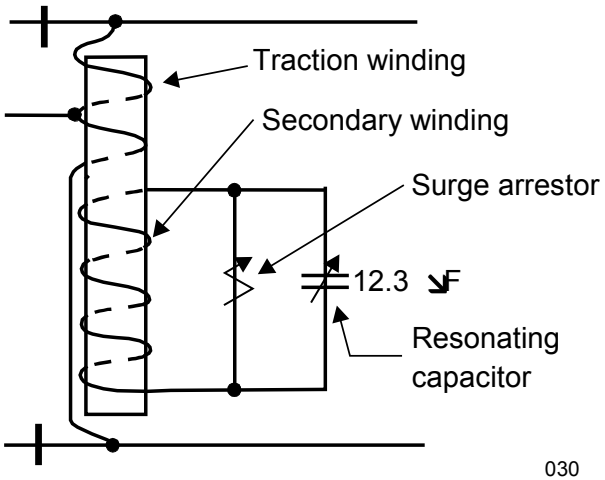



**Figure 25** 030

# Auto-coupled Impedance Bond

A further development of the resonated impedance bond is to use the secondary winding to carry the signalling current from and to the trackside equipment. The impedance bond is then said to be `auto-coupled'. Figure 26 shows an example of a double rail AC track circuit with auto-coupled impedance bonds used in a DC traction area. It also shows an intermediate resonated bond. In practice up to two intermediate resonated bonds per track circuit can be used for cross bonding to other lines or to substation busbars.

*Auto-coupled Impedance Bonds*

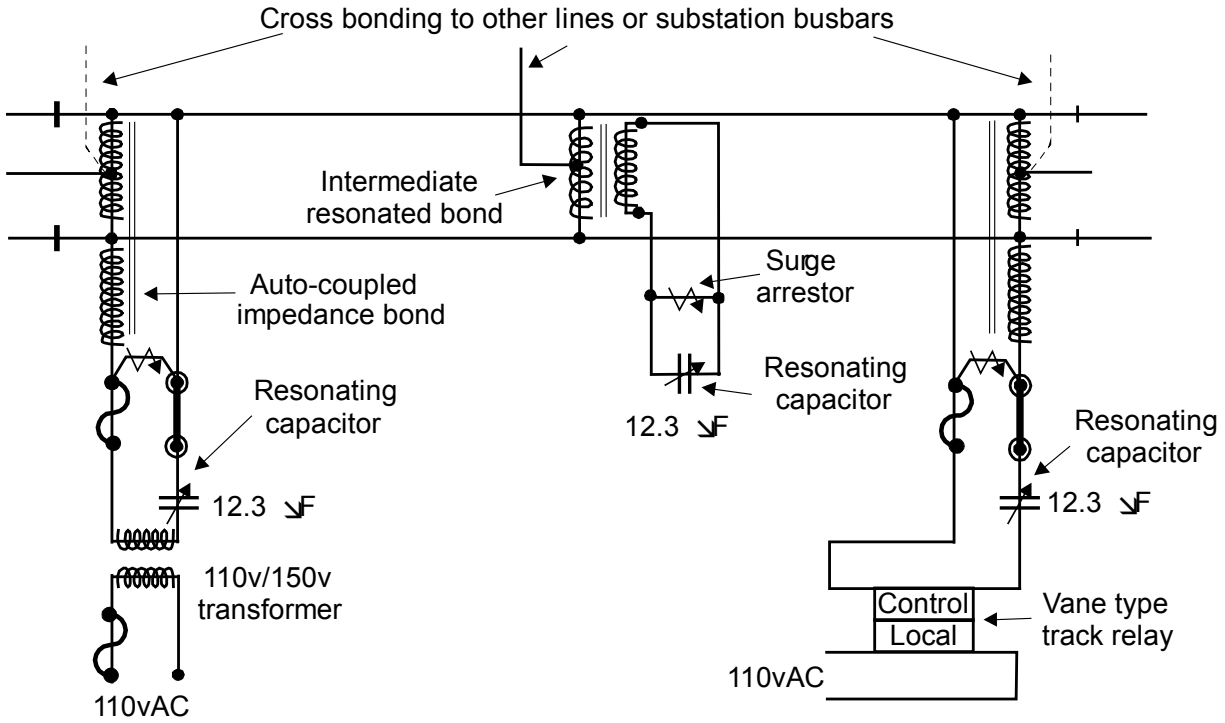



**Figure 26**

03

## Auto-coupled Impedance Bonds in AC Traction Areas

In some AC traction areas double rail traction returns are used in conjunction with AC track circuits. The traction supply is 50 Hz and the track circuit signalling is 83.3 Hz. Auto-coupled impedance bonds are used, resonating at 83.3 Hz. With the absence of DC and the traction return currents being much lower, the iron and copper sections in the impedance bonds are reduced and the air gap smaller. This produces much higher impedance and the turns ratio between the traction winding and the secondary winding is reduced to the order of 3:1.

# 7.3 Audio Frequency Track Circuits

In areas where DC and AC traction are used together and in non-electrified areas where long track circuits are required (such as in tunnels), audio frequency track circuits are used. These types of track circuits depend on the use of reed filters and are often called **reed track circuits**.

*Reed Track Circuits*

Reed filters and tuned circuits employ mechanically vibrating reeds which can be tuned with considerable precision and have a very high Q. (This is their ability to discriminate frequency differences.) The frequencies in common use for track circuit purposes are spaced only 3 Hz apart, for example 363, 366, 369, 372, 375 and 378 Hz.

*Audio Frequency Track Circuit Arrangement*

Figure 27 shows the simplified block diagram for an audio frequency track circuit.

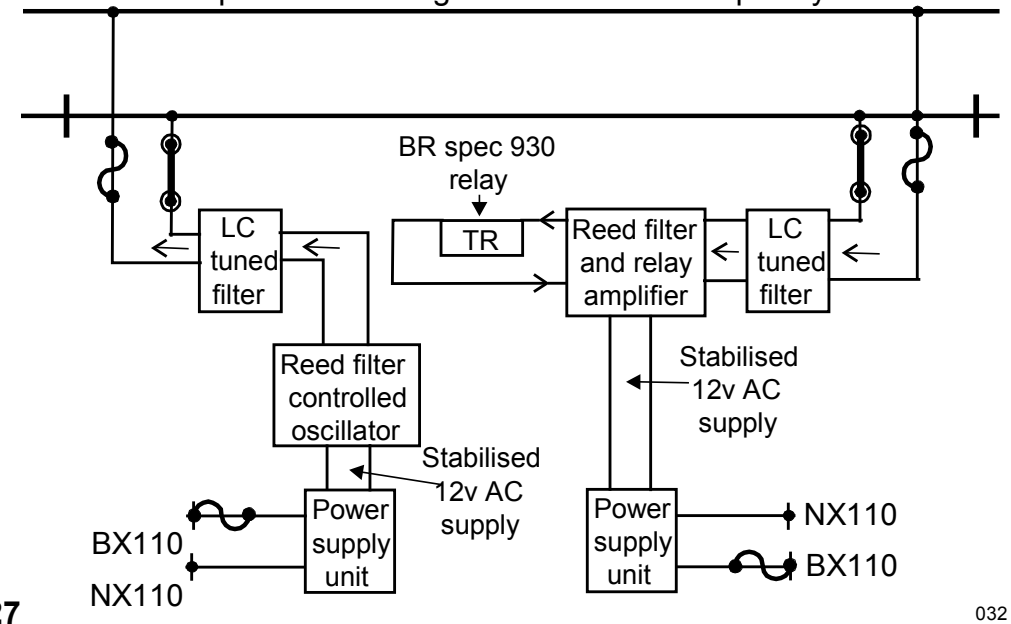



**Figure 27**

In figure 27 insulated rail joints mark the extremes of the track circuit. To avoid the effects of cross coupling and failure of the insulated joints, adjacent track circuits are energised at different frequencies. Normally no adjustments are provided and a minimum drop shunt resistance of 0.5 ohm is achieved for track circuits up to 2000 ft long with ballast resistances down to 6.0 ohms per 1000 ft. In special cases adjustments of the feed signal level can be provided to overcome very poor ballast conditions or for very long tunnels.

## Jointless Track Circuits

With continuously welded rail it is attractive to avoid the use of insulated rail joints. Jointless track circuits have been introduced to meet this need. They fall into two types:

- voltage operated
- current operated.

Figure 28 illustrates both types.

*Jointless Track Circuits*

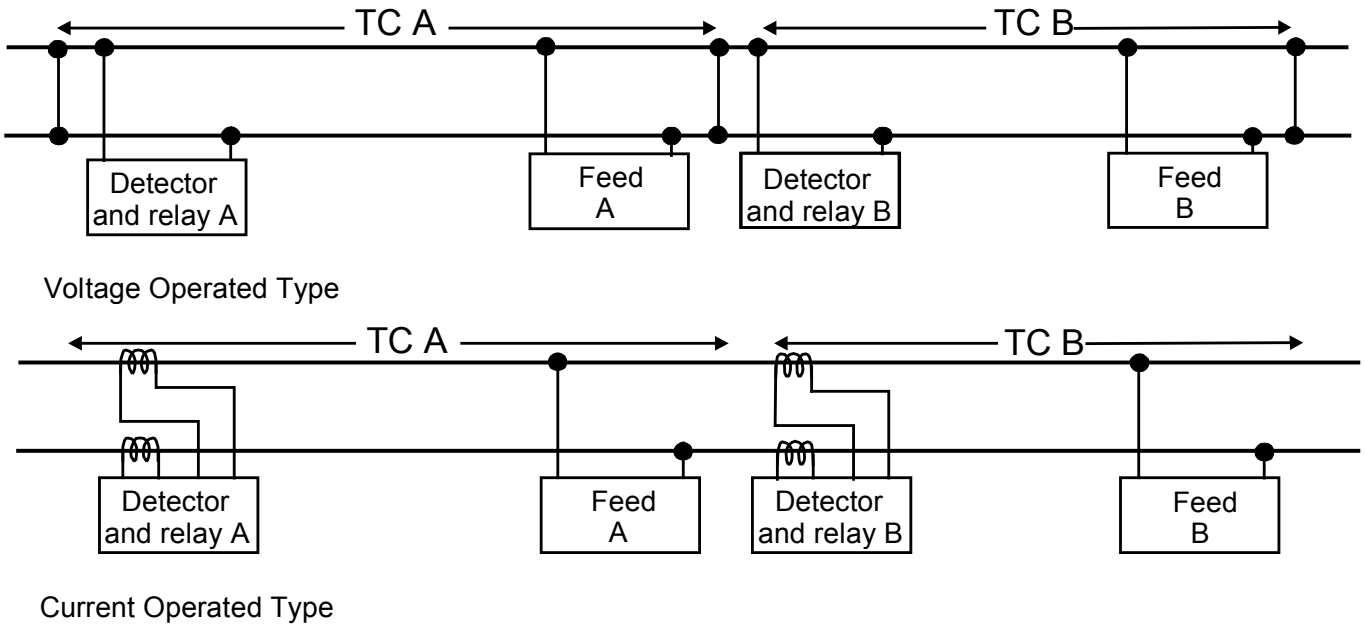



**Figure 28**

033

*Voltage Operated Type*

With the voltage operated type the ends of each track circuit are marked by short circuiting the two rails together. Provided the impedance of the short circuit is sufficiently low, very little of the energy from each feed crosses the boundary. The short circuit does not shunt the feed and detector/relay sets because the rails do have a significant impedance at the operating frequencies used, which are in the range 1500 to 3000 Hz. The presence of a train occupying the track circuit is detected because it substantially reduces the signalling voltage reaching the detector. The main weakness of the voltage operated type is that the train shunt impedance required to detect occupation decreases near the ends of the track circuit. This could lead to a wrong side failure. Practical arrangements involve increasing the impedances at the ends of the circuit to overcome this problem.

*Current Operated Type*

With the current operated type the ends of the track circuits are marked by coils located adjacent to each rail which are inductively coupled to the rail. Any alternating current flowing in the rail induces a voltage in the coil. When there is no train occupying the track circuit the current path is completed via the feed set of the adjacent track circuit and the ballast resistance. The presence of a train occupying the track circuit is detected when the current flowing in the rails is reduced and the voltage induced in the coils falls. There is no decrease in the train shunt impedance required to detect occupation at the ends of the track circuit, but there is another potential weakness when two trains occupy adjacent track circuits. If the train ahead (in TCB) has a higher shunt impedance than the train in rear (in TCA), the signalling voltage (in TCB) may be sufficient for the low shunt impedance of the train in rear to allow sufficient current to flow in the rails to cause the track circuit (TCB) to indicate clear. This will be a wrong side failure.

Practical examples of the two types of jointless track circuit are:

- voltage operated - Aster Type U
- current operated - Alsthom.

## Aster Type U Jointless Track Circuit

This is illustrated in figure 29.

*Aster Type U Jointless Track Circuit*

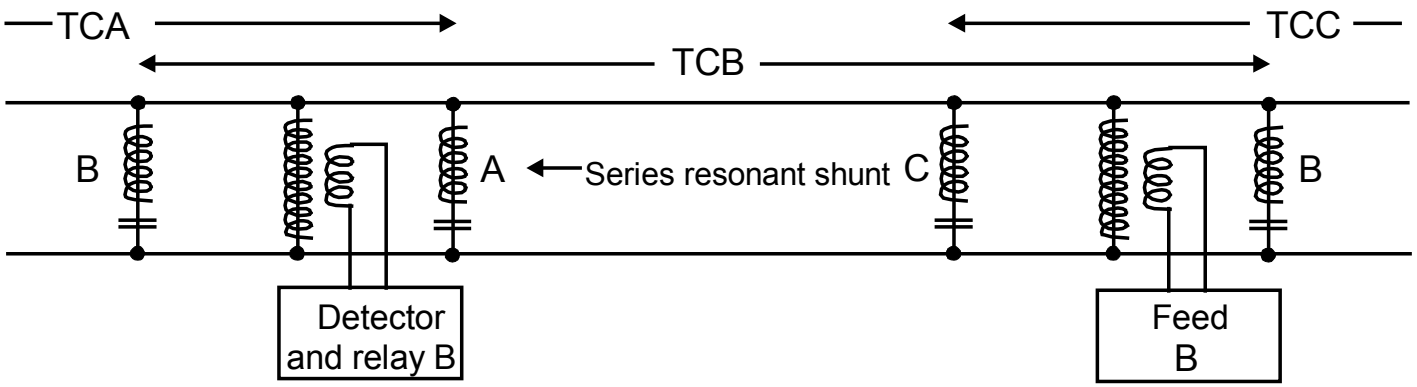



**Figure 29**

Each successive track circuit operates on a constant separate frequency in the band 1700 to 2600 Hz and its boundaries are marked by series resonant circuits tuned to that frequency. To overcome the problem of low shunt impedances occurring close to the ends of the track circuit, each track circuit is overlapped with the next. In the overlap the presence of a train should be detected by either or both of the circuits. When the track circuits are more than 1000 metres long the feed set is located at the middle with separate relay sets at both ends. This type of track circuit can be up to 2 km long with a minimum ballast impedance of 2 ohms per km.

## Alsthom Jointless Track Circuit

*Alsthom Jointless Track Circuit*

This type of track circuit has been designed specifically for use on electrified lines and is as illustrated for the current operated type in figure 28. It uses frequencies in a range between 300 Hz to 10 kHz and can operate over lengths up to 2 km with a minimum ballast impedance of 2 ohms per km. To obtain a clear distinction between the wanted and unwanted signals, the detector responds only to currents flowing in opposite directions in the two rails within the selected frequency slot. Unwanted signals derived from the traction supply tend to flow in the same direction in both rails and are rejected.

## 7.4 High Voltage Impulse (HVI) Track Circuits

This type of track circuit was specifically developed to deal with the problems of rails which are little used and prone to rust and other contamination. The high voltage pulse breaks down the rust film, etc.

Typical is the GEC-Alsthom HVI track circuit. A block diagram is shown in figure 30.

*HVI Track Circuit*

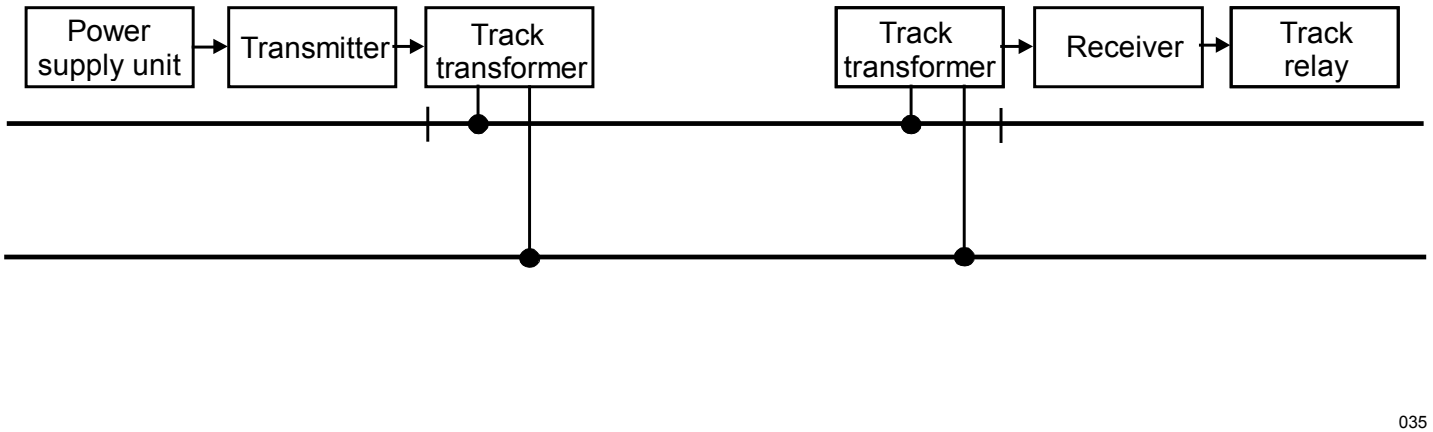



**Figure 30**

The HVI track circuit is suitable only for jointed rails. The pulse is generated by discharging a capacitor via a thyristor into the track transformer at a rate of 3 pulses per second. The pulse passed to the rails has an asymmetric shape as shown in figure 31.

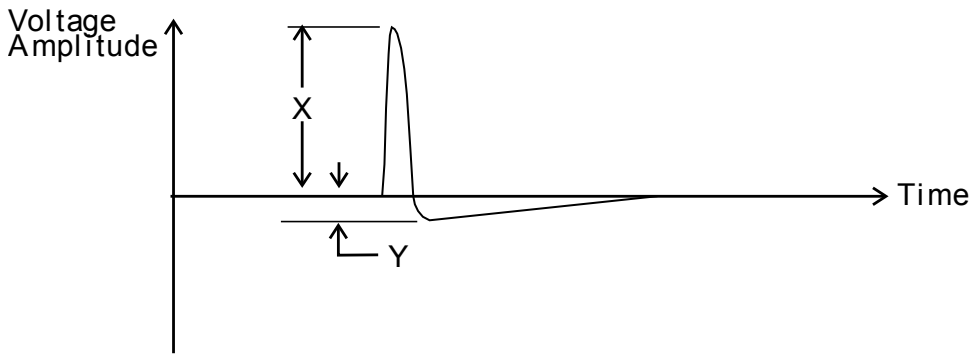



**Figure 31**

036

The normal amplitude ratio X:Y is 7:1. At the receiver the energy from the received pulse is rectified and directly drives the track relay. No power supply is required at the receiver. The positive and negative pulses are rectified separately and the relay picks only when their amplitudes are in the correct ratio.

This means that the relay response is polarity sensitive. For protection against IRJ failure it is normal practice to stagger the polarities of successive track circuits on the same line.

The HVI track circuit can be used on non-electrified lines and both DC and AC electrified lines. On electrified lines the track transformers have to be equipped with a series resistor to prevent the traction current from saturating or damaging them. Being transformer coupled the receiver is immune to DC. AC traction interference is rejected, because it does not continuously present the required asymmetrical wave form. Although they are potentially capable of operation over longer lengths of track, HVI track circuits are rarely used over more than 300 metres.

## 7.5 Section Summary

This section has looked at the main types of track circuits other than the DC type

- AC track circuits
- audio frequency track circuits
- high voltage impulse track circuits

Now complete Self-Assessment Questions 6 to 8 before proceeding to Section 8.

## Self-Assessment Question 6

Look at figure 32 and identify the type of track circuit illustrated. Then fill in the boxes with the correct names for the items indicated.

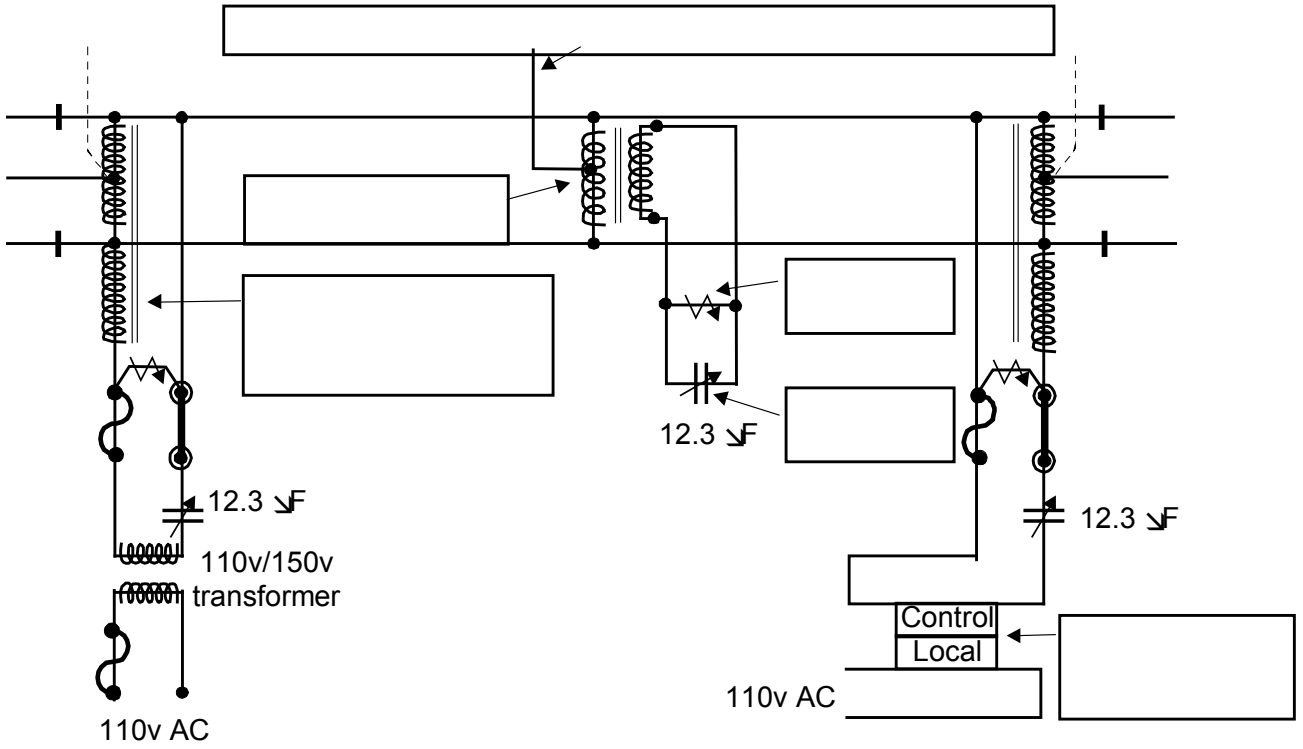



**Figure 32**

037

## Self-Assessment Question 7

Look at figure 33 and identify the type of track circuit illustrated. Then identify the function of the components labelled A, B, C, and D.

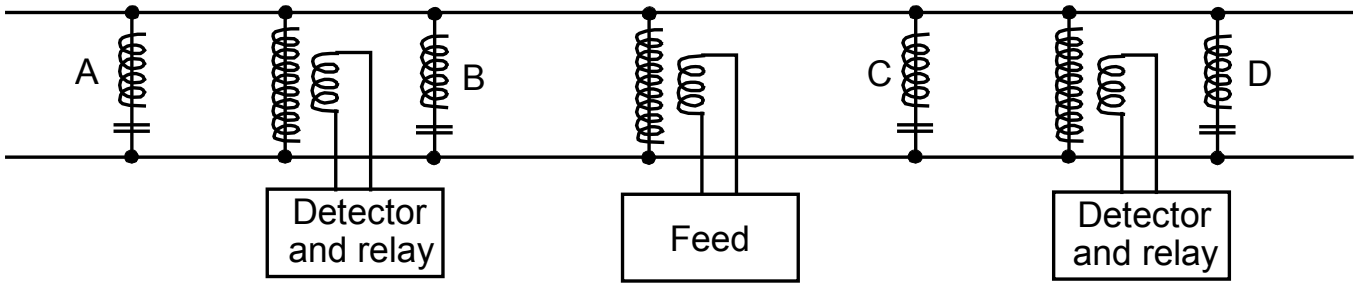
Type of Track Circuit:

A =

B =

C =

D =




038

**Figure 33**

## Self-Assessment Question 8

In what circumstances is it appropriate to use a high voltage impulse (HVI) track circuit? Is it suitable for use on non-jointed (continuously welded) rails? In electrified areas how does the HVI track circuit distinguish between traction generated noise and the track circuit pulses?

Now check your answer to Self-Assessment Question 6 to 8 with the ones given in the Answers Section. Before moving onto Section 8 make sure that you fully understand all the information given in Section 7. If there are any areas which you do not fully understand, then re-read the relevant parts of the text.

# 8 Module Summary

This module has looked at

- the track circuit
- the component parts of the DC track circuit
- the operation of the DC track circuit
- ballast resistance
- DC track configurations
- other types of track circuit

Before attempting the Self Test Questions which follow, make sure that you fully understand what you have read. Take your time and check your answers to the Self-Assessment Questions. When you feel confident begin the Self Test Questions.

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# 9 Self Test

Attempt the following Self Test Questions. Take your time and think about the answers carefully. When you have finished the Self Test, check your answers with those given in the Answers Section. If you have experienced any difficulties, return to the relevant part of the text and re-read the material. Do not move onto the next module until you feel confident in your understanding of this work.

## Self Test Question 1

State two functions of the track circuit.

## Self Test Question 2

Correctly identify the following components on figure 34.

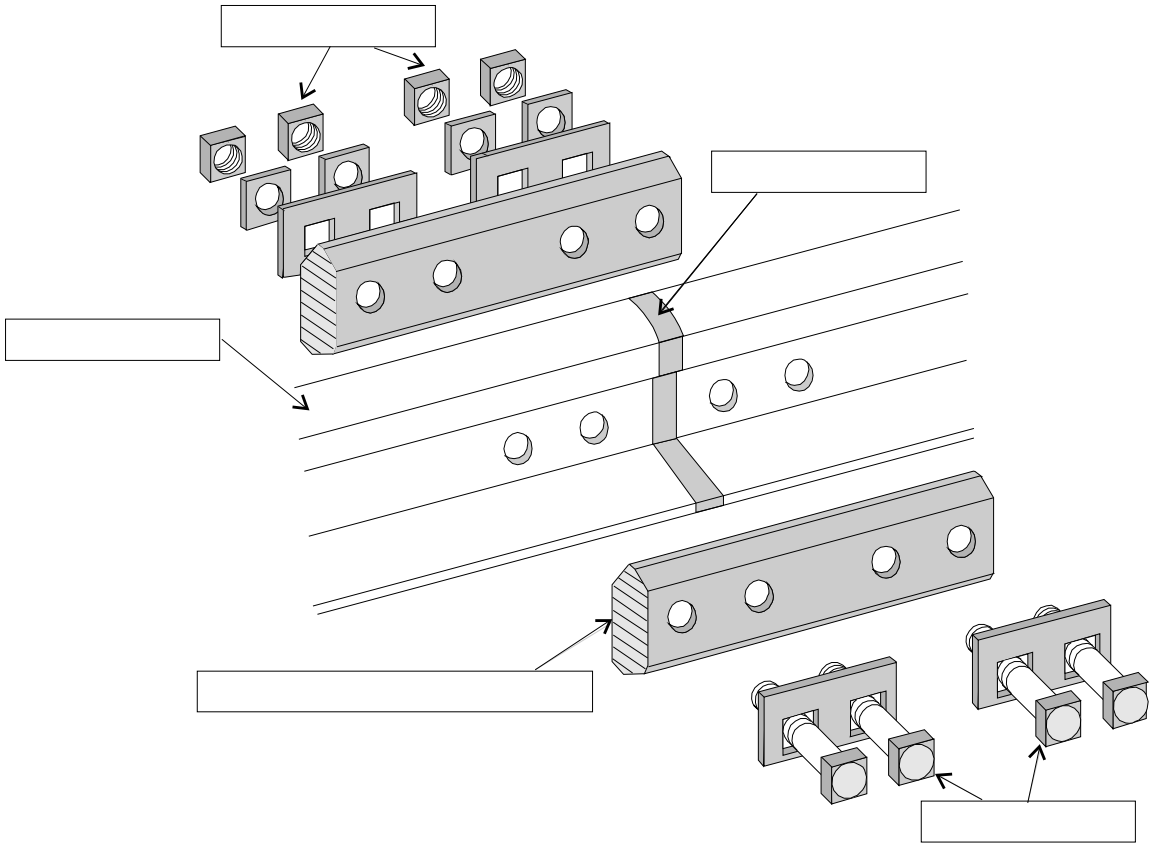



Figure 34

## Self Test Question 3

State the effect of wet weather on ballast resistance.

## Self Test Question 4

In your own words explain what the drop shunt value describes.

## Self Test Question 5

Correctly label the following figure which provides an example of a track circuit layout. State the condition which this circuit illustrates.

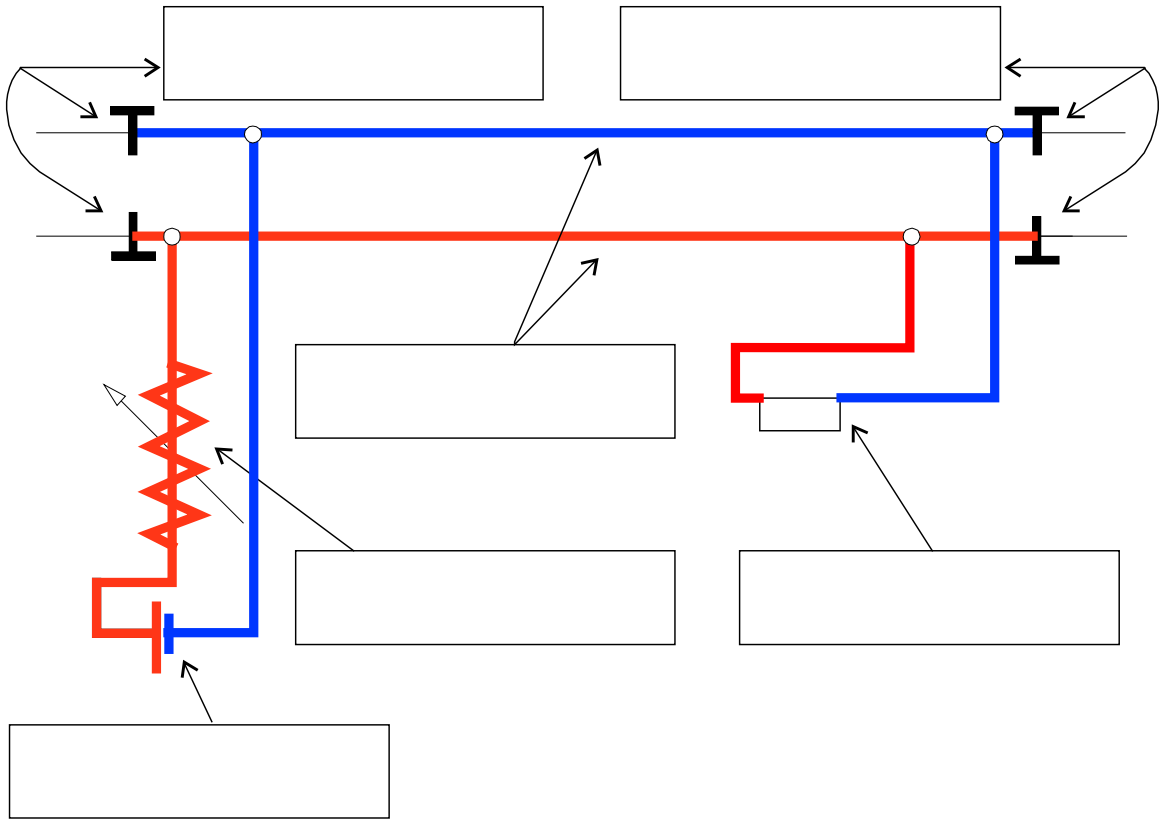


**Figure 35**

023

## Self Test Question 6

a) Is an impedance bond necessary when single rail operation is in use? Explain why

b) Is an impedance bond necessary when double rail operation is in use? Explain why.

c) Draw the arrangement of auto-coupled impedance bond at the feed end of an AC track circuit and label the components.

## Self Test Question 7

Look at figure 36. What type of track circuit arrangement does it illustrate? What are the distinctive features which you looked at to identify it?

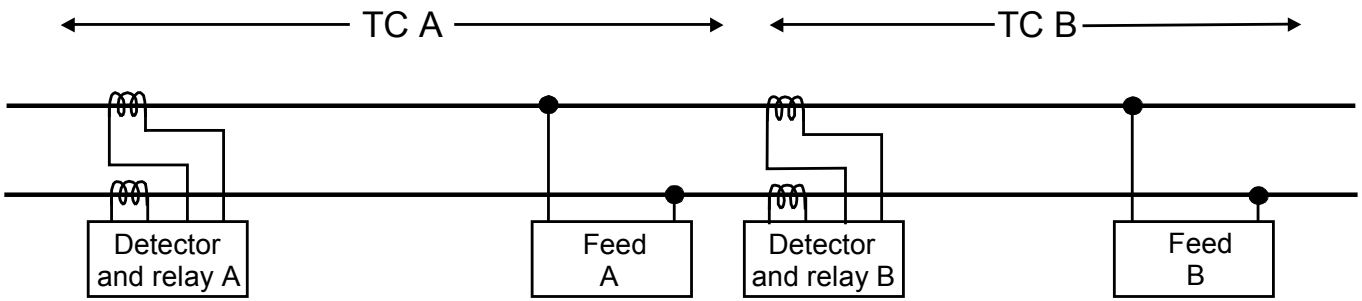



**Figure 36**

## Self Test Question 8

Draw the basic block diagram of a GEC-Alsthom HVI track circuit and the shape of the voltage pulse applied to the track.

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# 10 Answers

This section will provide the answers to both the

- Self-Assessment Questions
- Self Test Questions

Check your answers carefully. If you experienced any difficulties, re-read the relevant parts of the text.

# Answers to Self-Assessment Questions

## Answer to Self-Assessment Question 1

Figure 37 shows the component parts of the DC track circuit correctly labelled.

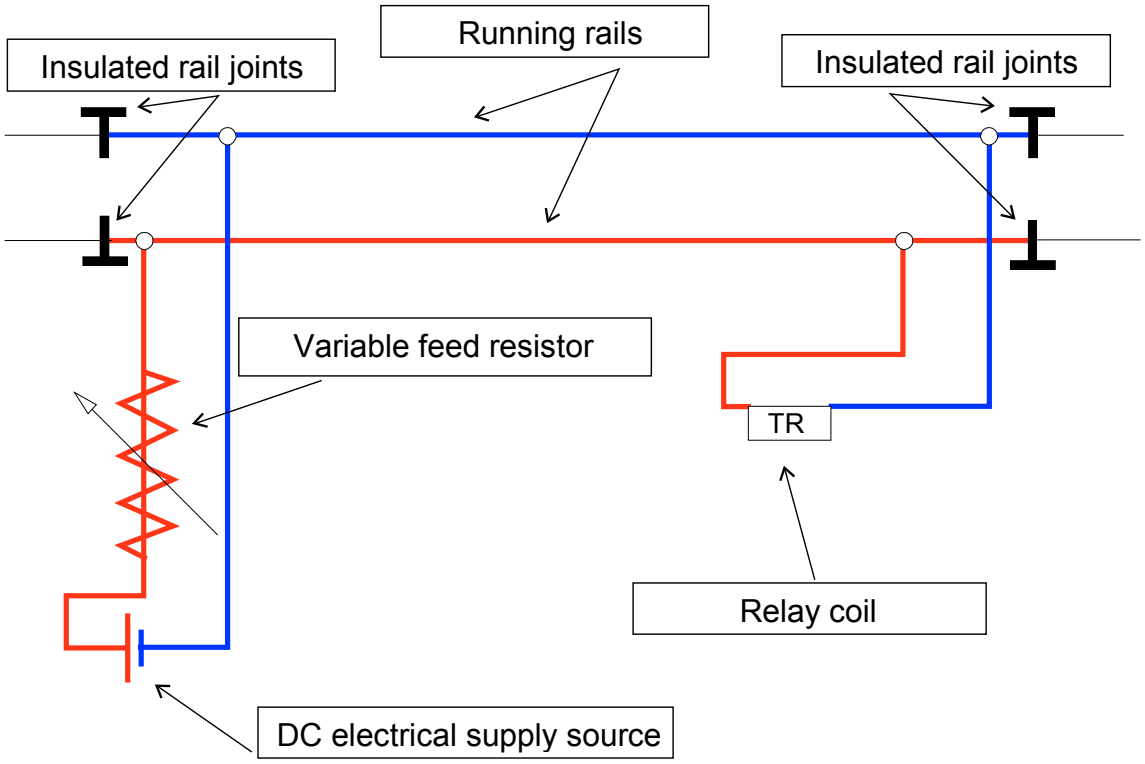



Figure 37

## Answer to Self-Assessment Question 2

The two basic conditions of the track circuit are

- track circuit unoccupied (clear)
- track circuit occupied

## Answer to Self-Assessment Question 3

Figure 38 shows a track circuit occupied

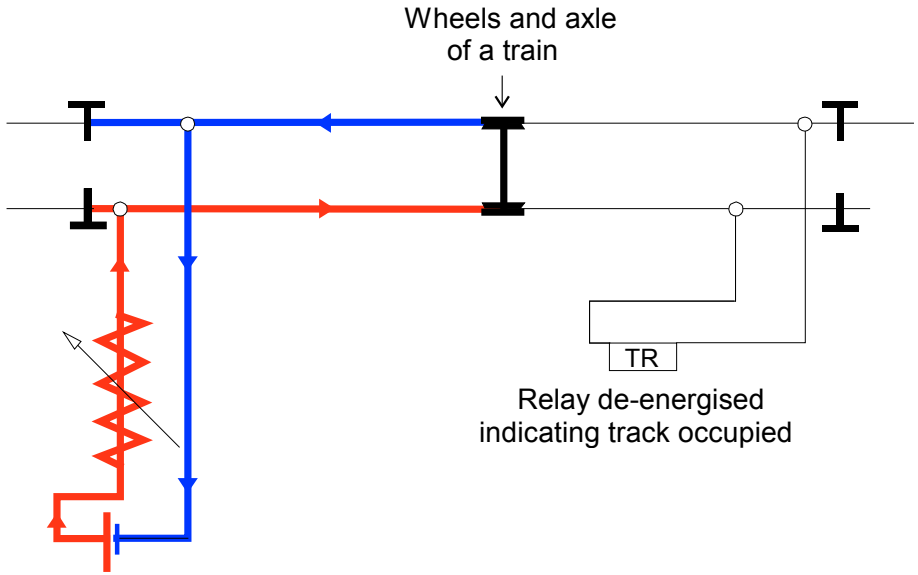



**Figure 38**

# Answer to Self-Assessment Question 4

| Ballast condition | Ballast resistance | Drop Shunt Value | Voltage at relay |
|---|---|---|---|
| Average | 10 ohms | 0.7 ohms | 0.6 ohms |
| Wet | **Less** | **More** | **Less** |
| Dry | **More** | **Less** | **More** |
| Frosty | **More** | **Less** | **More** |

**Table 2**

## Answers to Self-Assessment Question 5

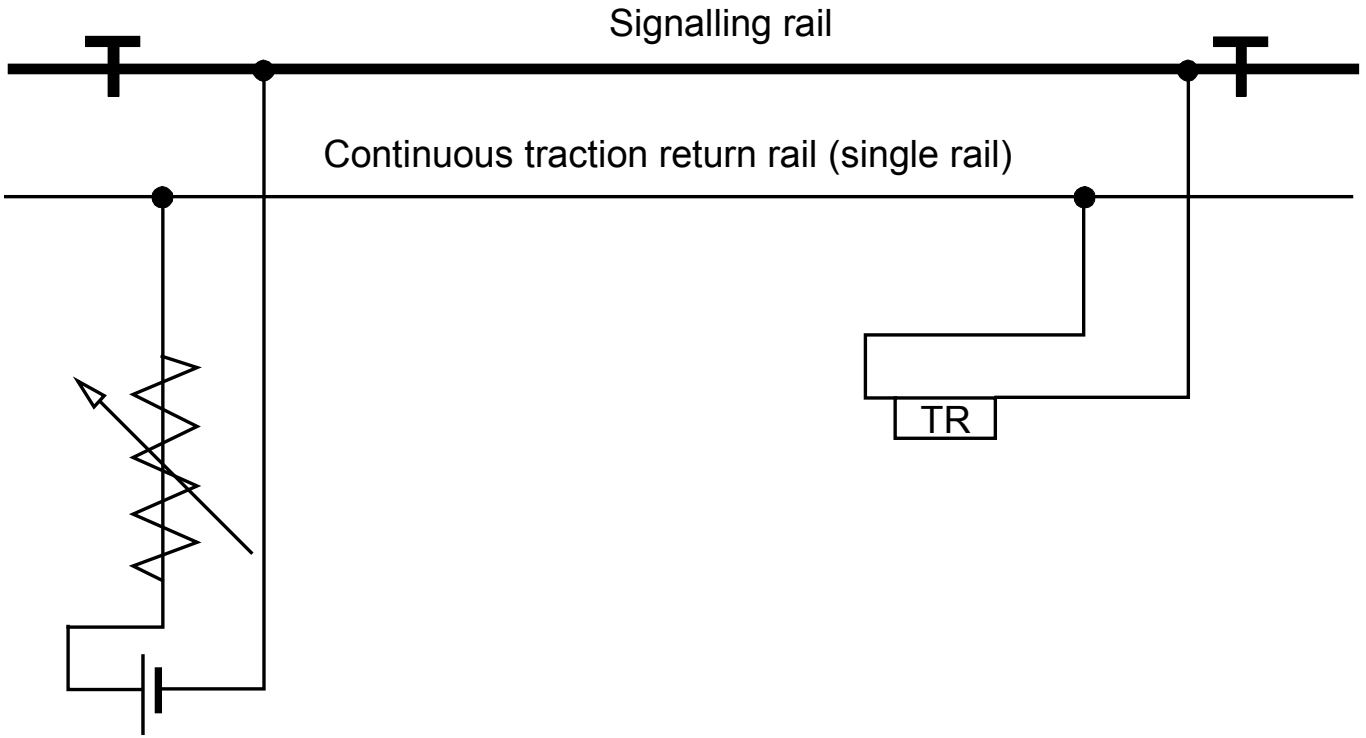
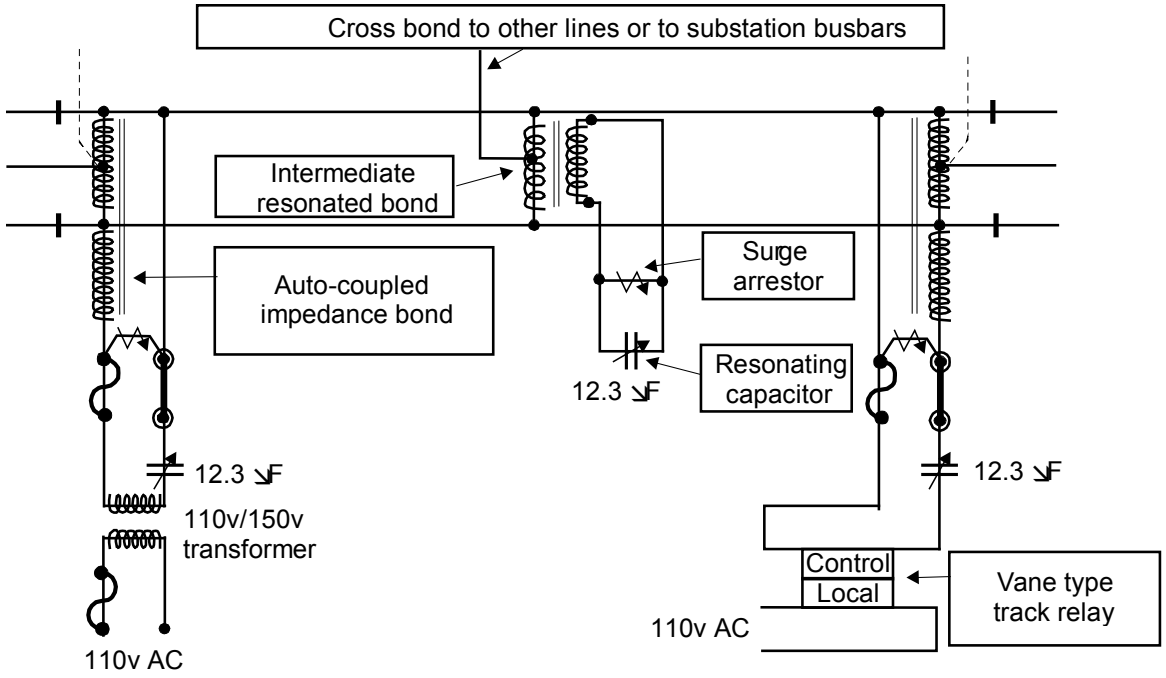



**Figure 39** 017

The track circuit configuration shown in figure 39 is an example of **single rail operation**

# Answer to Self-Assessment Question 6




This is ....
a double rail track circuit with resonated auto-coupled impedance bonds and an intermediate resonaded bond.

**Figure 40**

040

## Answer to Self-Assessment Question 7

Type of track circuit: Aster type U jointless track circuit with centre feed and two track relays.

A = Series resonant shunt marking end of track circuit in the middle of the figure.

B = Series resonant shunt marking end of track circuit to the left of the figure.

C = Series resonant shunt marking end of track circuit to the right of the figure.

D = Series resonant shunt marking end of track circuit in the middle of the figure.

## Answer to Self-Assessment Question 8

It is appropriate to use an HVI track circuit on lines which are little used and prone to rust and other contamination. It is **not** suitable for use on non-jointed rails. In electrified areas the HVI track circuit distinguishes between traction generated noise and the track circuit pulses by the following means

- the receiver is connected through the track transformer which de-couples it from DC
- the receiver responds only to continuous pulses of the correct polarity with the right ratio of positive and negative amplitudes
- AC traction noise tends to have symmetrical wave forms and rarely produces isolated wave shapes which match those of the HVI track circuit

# Answers to Self Test Questions

## Answer to Self Test Question 1

The functions of the track circuit are

- to prove that a section of track is clear of a train or vehicles
- to detect the presence of a train or vehicles
- to control signalling equipment

## Answer to Self Test Question 2

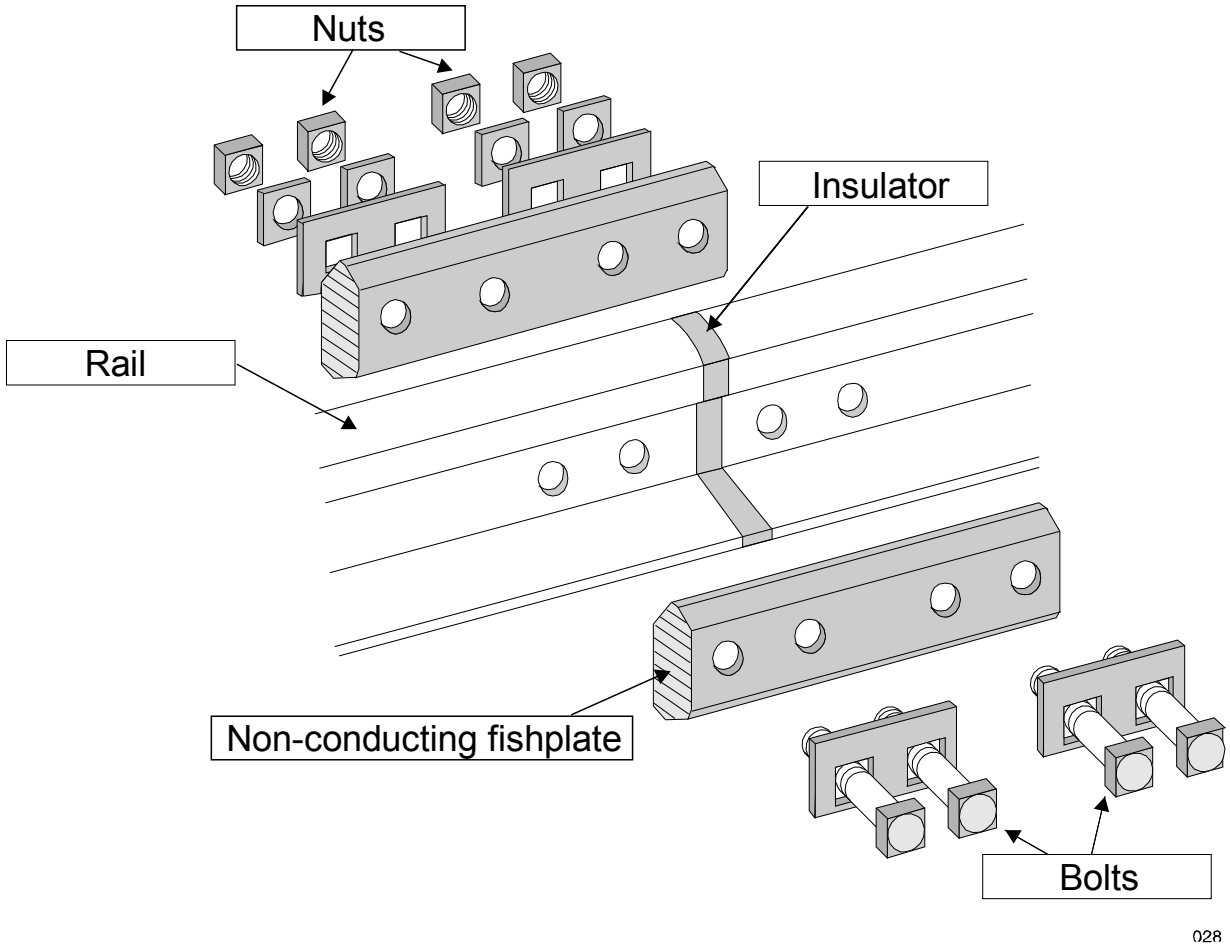



**Figure 41**

## Answer to Self Test Question 3

In wet conditions the ballast resistance is low. This means that more current is shunted away from the relay and the drop shunt value increases.

## Answer to Self Test Question 4

Drop shunt value is the maximum resistance which, when connected between rails at the relay end of the circuit, just causes the front contacts of the track relay to open.

## Answer to Self Test Question 5

Figure 42 provides a correctly labelled track circuit layout. The condition which the circuit illustrates is a **track circuit clear.**

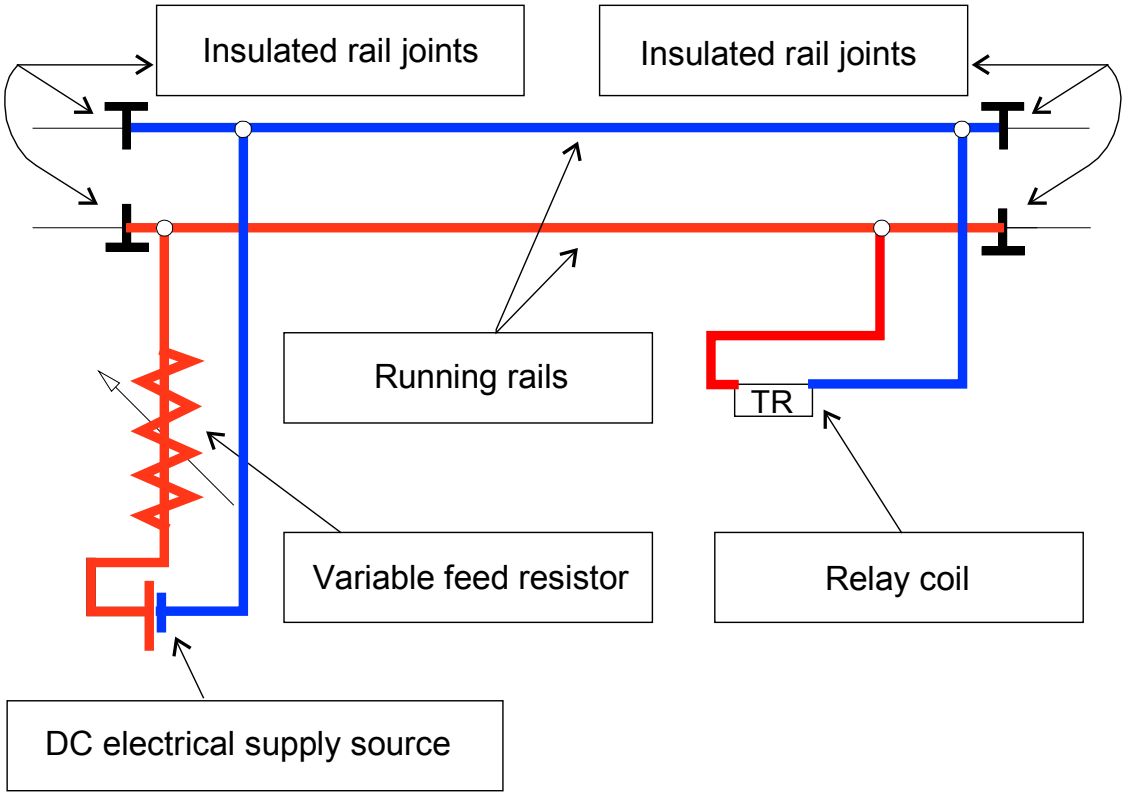


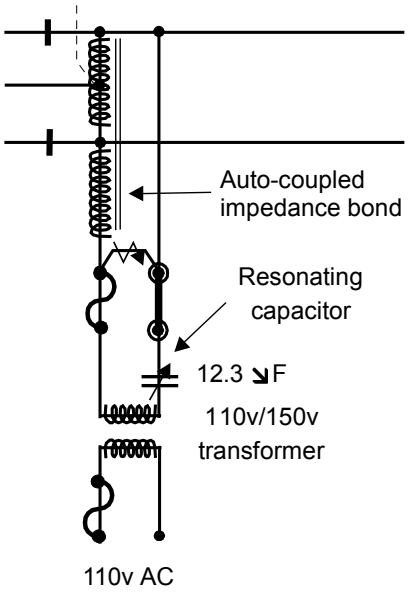

**Figure 42**

## Answer to Self Test Question 6

a) An impedance bond is not necessary when single rail operation is in use. This is because the traction current is flowing in one rail only and has no effect on the AC track circuit track side equipment.

b) An impedance bond is necessary when double rail operation is in use and an AC track circuit is being employed. This is because the AC track circuit requires the track to be fitted with insulated rail joints and a means has to be provided to shunt the joints for the returning traction current without preventing the track circuit from functioning.

c) The arrangement of the auto-coupled impedance bond at the feed end of an AC track circuit is shown in figure 43.




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**Figure 43**

## Answer to Self Test Question 7

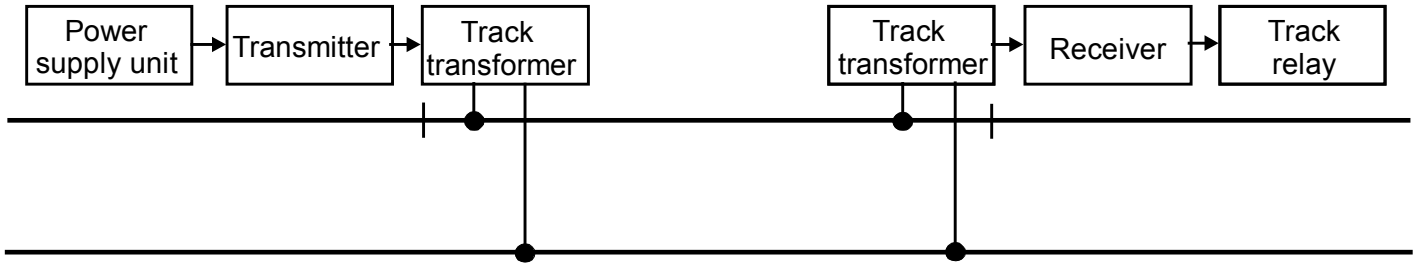
The track circuit illustrated is a current operated jointless track circuit. The distinctive features are

- no IRJs
- no resonant or other shunts connected between the rails
- the pick-up coils located on each rail at the detector/relay ends of the track circuits

## Answer to Self Test Question 8

The block diagram of a GEC-Alsthom HVI track circuit and the shape of the voltage pulse applied to the track are shown in figures 44 and 45.

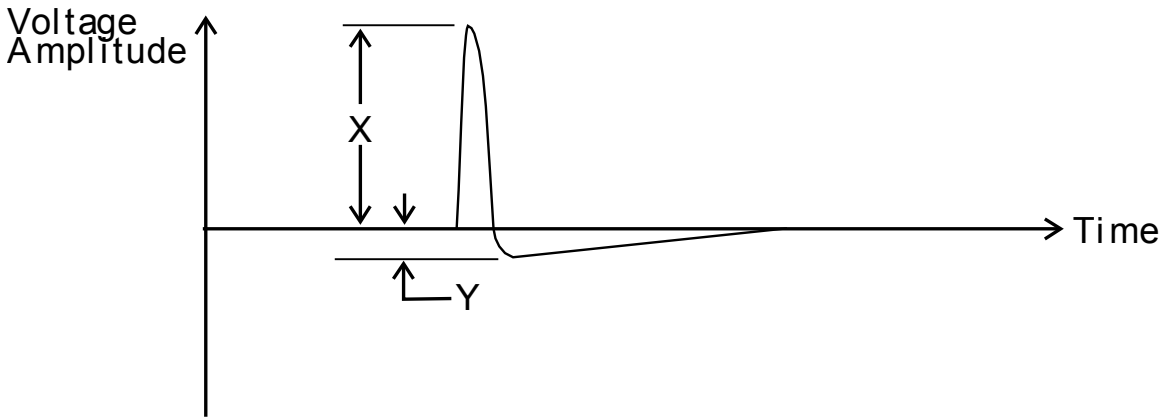
*Block Diagram for GEC-Alsthom HVI track circuit*




042

**Figure 44**

*Shape of Voltage Pulse*




043

**Figure 45**

# 11 Glossary

Within the railway system there are a number of business variations and individual conventions which mean that alternative terms may be used for

- equipment
- operations
- features

This module has used terms which normally correspond with those found in the Signalling Installation Handbook and the Track Circuit Handbook, however it may be useful for you to familiarise yourself with the alternative terminology which you may encounter in your work. This glossary provides a number of variations which may occur.

| | |
|---|---|
| Jumper wire | - Jumper, jumper bond |
| Drop shunt | - Drop away shunt, train shunt |
| Variable feed resistor | - Feed resistor, series resistor, regulating resistor, series regulating resistor |
| Insulated Rail Joint (IRJ) | - Insulated block joint |
| Signalling rail | - Signal rail, “our” rail |

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